REVIEW

Distribution shift, generalization and OOD challenge in offline reinforcement learning: a comprehensive survey

Ali Riahi Samani · Xujiang Zhao · Feng Chen

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Abstract
Offline Reinforcement Learning (RL) has emerged as a promising paradigm to overcome the limitations of conventional RL methods that depend on extensive and often unsafe online interactions with the environment. By learning policies exclusively from pre-collected datasets, Offline RL aligns with the growing emphasis on data efficiency and safety in machine learning. However, existing off-policy RL algorithms encounter substantial difficulties when trained purely on offline data, primarily due to distributional shift between the training dataset and the learned policy. This issue becomes more pronounced with high-dimensional function approximation, leading to degraded performance and poor generalization. This survey provides a comprehensive overview of recent advances in Offline RL, with a particular focus on the challenges of distribution shift, generalization, and out-of-distribution (OOD) actions. We review the evolution of this field, discuss data limitations, and analyze the historical and theoretical foundations of distributional shift in Offline RL. Furthermore, we categorize existing approaches into four major groups including Q-value restriction methods, uncertainty-based Q-restriction methods, policy constraint methods, and uncertainty-based policy constraint methods, highlighting their core principles and practical implications. We also summarize common benchmarks such as D4RL, discuss dataset enhancement strategies, and examine evaluation metrics and computational efficiency. Through this analysis, the survey offers a unified perspective on current solutions and identifies open challenges, providing valuable guidance for advancing robust, generalizable, and trustworthy Offline RL in domains such as robotics, healthcare, and autonomous systems.

Keywords Reinforcement learning · Offline RL · Distribution shift · Out of distribution · Uncertainty quantification · Multi-task data sharing

1 Introduction

Reinforcement Learning (RL) is a method in which an artificial intelligence (AI) agent learns optimal behavior through trial and error, guided by reward feedback from the environment [1]. RL has been particularly effective in solving complex sequential decision-making tasks, especially in environments where building explicit models is either impractical or costly [2]. However, traditional RL methods face significant limitations when continuous interaction with the environment is not feasible, leading to the rise of Offline RL. Offline RL leverages large

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datasets from prior interactions, allowing agents to learn without the need for additional online engagement. Yet, this approach introduces its own set of challenges, including reliance on static data, the potential for distributional shifts, and the absence of exploratory actions that are central to traditional RL [3].

Deep Reinforcement Learning (DRL), a specialized form of RL, integrates deep neural networks to approximate functions such as value functions, policies, and reward models [4]. While DRL has significantly advanced the capabilities of RL, it requires vast amounts of data, which poses challenges in real-world applications where data acquisition can be costly or limited. The reliance on large datasets further complicates the training process and can lead to issues such as distributional shifts, especially in Offline RL scenarios, where training is conducted using only previously collected data [5, 6]. The complexities are further amplified by high-dimensional function approximation, intensifying the pivotal issue of distributional shift inherent in offline RL.

The field is actively advancing to improve the adaptability of offline RL algorithms across diverse and complex scenarios, thereby broadening their practical use [7]. A key area of focus is the reliance on existing datasets, emphasizing the need for efficient learning from limited and possibly flawed data. This challenge is heightened by the static nature of training data and the absence of exploratory actions during learning, which are essential in traditional RL settings. These issues restrict the ability to explore and find potentially beneficial actions not present in the dataset [8]. Additionally, offline RL deals with counterfactual queries, complicating the evaluation of algorithms trained on specific datasets under different conditions [9].

To overcome these challenges, researchers have developed various solutions, including model ensembles, applying uncertainty penalties to value functions, and diverse policy regularization techniques. Methods such as pessimistic initialization [10], density ratio-based method [11], and conservative Q-learning [12] help address issues like distributional shift, out-of-distribution (OOD) actions, and overestimation bias, leading to more robust offline RL frameworks. These solutions involve enhancing existing RL algorithms, introducing new training paradigms, and integrating advanced deep learning techniques to improve policy learning. Tailoring offline RL strategies for specific uses, such as robotics, requires overcoming adaptability challenges and compensating for limitations in training data.

This study explores these challenges and the methodologies designed to address them, highlighting their impact on offline RL. We present a structured classification of techniques for tackling distributional shifts, complete with a detailed analysis of their algorithmic innovations, advantages, and limitations. This helps researchers choose the most appropriate algorithms for their offline RL projects. Lastly, we share our thoughts on unresolved issues and suggest potential directions for future research in this rapidly progressing field. Our contributions include:

1. Performing an extensive analysis of existing surveys to uncover research gaps,
2. Classifying methodologies into four unique categories to cover a wide range of approaches in distribution shift and OOD in offline RL,
3. Compilation of an exhaustive list of benchmarks for evaluating Offline RL algorithms, enabling more effective comparison and validation of methods,
4. Evaluation metrics analysis to assess the effectiveness of Offline RL techniques in diverse real-world scenarios,
5. Proposing future directions for the field, highlighting unresolved issues and suggesting innovative approaches to overcome existing limitations.

The rest of paper is organized as follow. In Sect. 2, previous surveys in the field are reviewed. Reinforcement learning concept are explained in Sect. 3. Data limitations and distributional shift are examined in Sects. 4 and 5. In Sect. 6 classification of existing methods is provided. Benchmarks are introduced in Sect. 7. Section 8 analyze evaluation metrics. Section 9 delves into the existing challenges. A thorough discution is provided in Sects. 10 and 11 concludes the study.

2 Related surveys

Various studies have examined the challenge of distribution shifts as well as OOD and have conducted comprehensive surveys on this issue. Wu et al. [13] conducted a thorough examination of how graph learning methods can effectively manage distribution shifts. Their primary objective was to consolidate and summarize the latest strategies and insights in handling these shifts within graph-based learning paradigms. They categorized their investigation into three main areas: graph domain adaptation learning, focusing on techniques to adapt graph models to new domains with varying data distributions; graph OOD learning, which addresses methods for handling data significantly different from the training distribution; and graph continual learning, where they explored strategies to update models continuously as new data arrives without forgetting previously learned information. Their contributions include a comprehensive taxonomy of existing approaches based on the observability of distribution shifts during inference and the availability of supervision during training. They also identified key challenges such as the dynamic nature of real-world graph data and outlined future research directions to enhance model robustness under distribution shifts.

Li et al. [14] conducted a comprehensive review focused on methods and strategies for addressing OOD generalization challenges in graph learning. Their primary objective was to systematically summarize the latest advancements in this field. They began by formally defining OOD generalization on graphs, highlighting the difficulties arising from distribution shifts between training and testing datasets. The authors categorized existing approaches into three main classes: data-level methods, which involve techniques like data augmentation to improve generalization; model-level methods, which modify model architectures or learning algorithms for enhanced robustness; and learning strategy-level methods, which encompass different training paradigms and optimization techniques tailored for OOD scenarios.

Yu et al. [15] systematically summarized existing research on methods for evaluating OOD generalization in machine learning models. Their primary goal was to provide insights crucial for deploying models in real-world applications where robust performance on unforeseen data is essential. They began by defining the challenges posed by OOD generalization, particularly the shifts in data distribution between training and testing phases. The authors categorized evaluation methods into three main paradigms: OOD performance testing, which assesses model performance on OOD data directly; OOD performance prediction, which predicts model behavior on OOD data without actual testing; and OOD intrinsic property characterization, which examines inherent model properties influencing OOD generalization.

In other studies, Liu et al. [16] assessed the methodologies for OOD generalization and categorized them into three segments: unsupervised representation learning, supervised model learning, and optimization, based on their roles within the overall learning process. Yang et al. [17] reviewed the technical advancements in OOD detection and provided a survey introducing a unified framework called generalized OOD detection, which includes the five key issues: anomaly detection, novelty detection, open set recognition, OOD detection, and outlier detection. Ghassemi and Fazl-Ersi [18] analyzed over 70 papers in this field, highlighted challenges, and suggested directions for future research. Their work provided a comprehensive perspective on different types of data shifts and proposes solutions for improved generalization.

Cui and Wang [19] discussed fundamental aspects of OOD detection using deep learning and classified methods based on the type of training data. They divided OOD detection into three categories: supervised, semi-supervised, and unsupervised. For supervised data, they further categorized the methods into model-based, distance-based, and density-based approaches according to the technical methods used. Currently, there are few comprehensive reviews in the realm of offline RL.

A significant contribution is the tutorial article by Levine et al. [20], which outlines fundamental challenges, examines groundbreaking research, explores practical applications, and considers unresolved issues within the field. It outlines the major challenges in applying offline learning and dynamic programming methods in the fully offline setting, including high variance in importance sampling, distributional shift issues, and the difficulty of

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obtaining calibrated uncertainty estimates. It also highlights open problems in policy constraint methods and model-based RL approaches in the offline setting. The work suggests that policy constraint methods could mitigate these issues by ensuring actions used for computing target values are close to the behavior distribution, thereby avoiding OOD queries to the Q-function.

Another survey in offline RL is done by Figueiredo Prudencio et al. [6] which introduces a new taxonomy that uses a standardized notation to classify the various components and processes that can be assembled to create an offline RL algorithm. It suggests that offline RL methods can be categorized into diverse groups such as model-based, one-step, and imitation learning methods. Additionally, it assesses the existing benchmarks in the field, examining the extent to which their datasets exhibit the essential characteristics desired in offline RL datasets, as well as analyzing the performance of individual methods and categories of methods relative to each dataset property.

Chen et al. [21] reviews current research to highlight the potential advantages and obstacles of offline RL in recommender systems. Unlike traditional RL methods that require continuous online interactions, offline RL utilizes existing datasets, offering a cost-effective alternative for systems with substantial offline data. The authors explore benefits such as enhanced data efficiency and scalability, alongside challenges like ensuring data quality and deploying learned policies effectively in real-world scenarios.

In this study, we focus exclusively on methodologies that address distribution shifts and OOD within the context of offline RL—an area that remains underexplored in prior surveys. Table 1 present the focus of previous work. Our work distinguishes itself in several critical ways:

1. Focused Scope: While many related surveys, such as Wu et al. [13], Li et al. [14], and Yu et al. [15], concentrate on OOD challenges within graph learning or general machine learning paradigms, they do not delve into the unique complexities of offline RL. Foundational studies like those of Levine et al. [20] and Figueiredo Prudencio et al. [6] provide valuable insights into offline RL but lack an in-depth examination of recent advances, particularly in tackling distributional shifts and OOD issues from the past four years. Our work bridges this gap by systematically analyzing these recent developments, offering a targeted review tailored to offline RL.
2. Comprehensive Categorization: Unlike earlier surveys that broadly address offline RL methods, we introduce a detailed categorization focused on Q-value restriction and policy constraint approaches, with a particular emphasis on the critical role of uncertainty quantification. This structured framework not only organizes the existing methodologies but also provides actionable guidance for researchers and practitioners in selecting the most suitable algorithms based on their specific requirements.
3. Exhaustive Methodological Analysis: Our survey goes beyond the broader overviews presented in previous works, such as Figueiredo Prudencio et al. [6], by offering an in-depth examination of algorithmic innovations, strengths, and limitations of methods specifically designed to address distributional shifts and OOD in offline RL. This level of detail ensures a clearer understanding of the trade-offs and practical considerations involved in adopting these approaches.

Table 1 Objective of related works

Refs.	Year	Objective
[20]	2020	Offline RL
[14]	2022	OOD generalization challenges in graph learning
[18]	2022	OOD Detection
[19]	2022	OOD Detection based on deep learning
[16]	2023	OOD generalization methods
[6]	2024	Offline RL
[21]	2024	Offline RL within recommender systems
[13]	2024	Distribution shifts in graph learning methods
[15]	2024	Evaluating OOD generalization
[17]	2024	Generalized OOD Detection
Ours	–	Distribution shift and OOD in offline RL

By addressing these critical gaps, our study not only offers a comprehensive and up-to-date review but also establishes itself as a key resource for advancing the understanding and development of robust, generalizable offline RL solutions.

3 Reinforcement learning concept

RL is a computational framework wherein agents are trained to optimize predefined reward functions through the development of policies, which delineate optimal actions within diverse domains [5].

3.1 Markov decision process

RL constitutes a specialized branch of machine learning that concentrates on devising algorithms capable of controlling dynamical systems, typically conceptualized as Markov Decision Processes (MDPs). The primary objective of RL is to formulate optimal strategies or policies that maximize the aggregate rewards within a specified environment.

An MDP, the quintessential model underpinning RL, is characterized by a tuple $M = (S, A, T, d_0, r, \gamma)$ where S denotes the set of states $(s \in S)$, encapsulating the feasible configurations of the environment, which may be either discrete or continuous. A represents the set of actions $(a \in A)$ available to the agent, which similarly can be categorized as discrete or continuous. The transition probability function $T(s_{t+1}|s_t, a_t)$ specifies the probability of transitioning to a subsequent state s_{t+1} given the current state s_t and action a_t. The initial state distribution $d_0(s_0)$ outlines the probability distribution across initial states. The reward function $r : S \times A \rightarrow \mathbb{R}$ assigns rewards to state-action pairs, and γ denotes the discount factor $(0 < \gamma \leq 1)$, employed to diminish the value of future rewards, thereby ensuring the summation of rewards converges in scenarios with infinite horizons [22].

For situations where the agent lacks complete observability of the state, the MDP framework is augmented to a Partially Observable Markov Decision Process (POMDP), described by $M = (S, A, O, T, d_0, E, r, \gamma)$. Here, O represents the set of observations $(o \in O)$, and E is the emission function $E(o_t|s_t)$, which articulates the likelihood of perceiving observation o_t given the state s_t. This extension accommodates scenarios where direct state information is inaccessible, necessitating inference based on observations to deduce the underlying state [23].

3.2 Reinforcement learning formulation and objective

In RL [5], the learning paradigm is centered around an agent's interactions with the environment, guided by a behavior policy. This process entails the agent's observation of states, selection of actions based on these states, and receipt of feedback through state transitions and rewards, which inform the iterative refinement of its policy. A policy, denoted as $\pi(a_t|s_t)$, encapsulates the agent's strategy, defining a conditional probability distribution over actions given the current state. In the context of POMDPs, the policy may additionally rely on current observations $\pi(a_t|o_t)$ or a history of observations $\pi(a_t|o_{0:t})$. The policy's evolution is predicated on the transitions (s_t, a_t, s_{t+1}, r_t) observed during the agent's environmental engagement, with the learning dataset D potentially encompassing a complete or partial collection of these transitions.

A trajectory τ, representing a sequence of states and actions across a defined horizon H, is formalized as $\tau = (s_0, a_0, \ldots, s_H, a_H)$. The distribution of such trajectories under a given policy π and an MDP M is expressed as:

$$\mathbb{P}_\pi(\tau) = d_0(s_0) \prod_{t=0}^{H} \pi(a_t \mid s_t) T(s_{t+1} \mid s_t, a_t) \tag{1}$$

The agent's objective, $V(\pi)$, aims to maximize the expected sum of discounted rewards along a trajectory, reflecting the cumulative reward. This objective is formulated as:

Table 2 Comparing online and offline RL

Parameter	Online RL	Offline RL
Data collection	Continuously gathers new data through ongoing interactions	Relies on pre-collected, static datasets for training
Safety and cost	can be costly and hazardous in domains like robotics and healthcare	Avoids risks by using existing data, making it safer and more cost-effective
Training Efficiency	regularly updates policies with fresh data but may face challenges in complex or expensive environments	Can utilize extensive datasets for improved generalization, though it struggles with issues like distributional shift, which arises from relying solely on static data

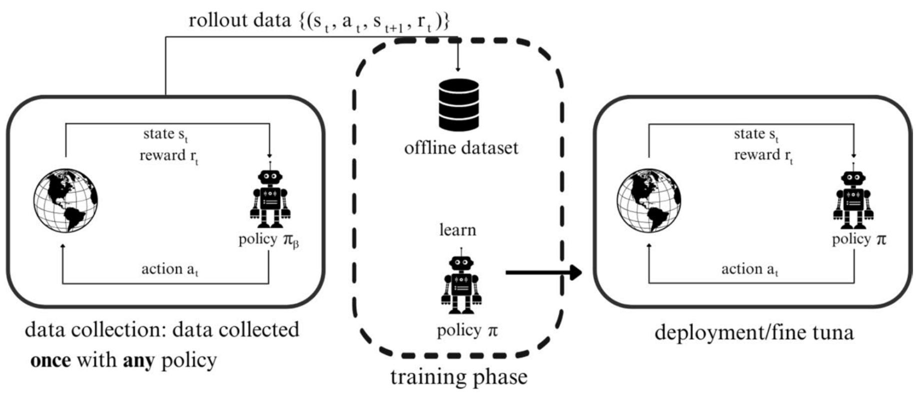


Fig. 1 Offline reinforcement learning

$$V(\pi) = \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{H} \gamma^t r(s_t, a_t) \right] \tag{2}$$

where H might extend to infinity, encapsulating the long-term reward maximization goal of the agent. The state visitation frequencies, both overall $d_\pi(s)$ and at specific timesteps $d_{\pi_t}(s_t)$, are derived from the trajectory distribution, underpinning the RL objective's foundation.

3.3 Online vs. offline RL

Online RL involves a dynamic process where an agent continually interacts with its environment, gathering new experiences to update its policy iteratively [24]. This approach allows the agent to learn and improve in real-time, utilizing the latest policy for decision-making. However, the necessity for constant interaction poses significant challenges in many applications, such as robotics, healthcare, and autonomous driving, where data collection can be expensive or risky [20].

Offline RL, also known as batch RL, offers an alternative by learning from pre-existing datasets of collected experiences. Instead of continually interacting with the environment, offline RL uses static datasets to train the policy [25]. This approach can leverage large amounts of previously gathered data to develop effective decision-making policies without additional interactions [26].

Online RL is capable of real-time learning and adaptation, making it highly effective in environments where continuous interaction is feasible. Offline RL is more practical in high-risk or costly environments and can leverage large datasets to train effective policies. Table 2 present key differences of online and offline RL.

3.4 Offline RL strategies

Figure 1 depicts the structure of offline RL, where an agent learns a policy from pre-collected data without interacting with the environment during training. In the **data collection phase**, an agent governed by a behavior policy π_β interacts with the environment, generating experience tuples $\{(s_t, a_t, s_{t+1}, r_t)\}$ that are stored in an offline dataset. The **training phase** then uses this fixed dataset to learn a new policy π, without further environment access. Finally, in the **deployment or fine-tuning phase**, the learned policy π is applied to the real environment and optionally refined through additional online interaction.

Unlike traditional online RL, which learns through continuous interaction with the environment, offline RL learns solely from a fixed dataset, similar to supervised learning. The terminology in this domain varies: "off-policy RL" refers to algorithms that learn from transition datasets collected under different policies, while “batch RL” denotes methods that use data batches for updates. In this survey, we adopt the term “offline RL” to avoid ambiguity. Theoretically, any off-policy RL algorithm might be adapted for offline use, with rudimentary approaches [27] possibly employing Q-learning with dataset D for initialization. However, the efficacy of such methods can be inconsistent, often necessitating subsequent online refinement.

Offline RL encompasses both model-based strategies, which either develop or employ models of the environment [28], and model-free approaches, which directly ascertain policies or value functions. Model-based methods are valued for their sample efficiency, enabled by the ability to simulate future states, but their performance depends on the accuracy of the environmental models. These approaches are especially relevant in areas such as robotics, where data collection is slow and risky. In offline RL, the focus is on developing robust agents from large datasets without interacting with the environment, with later online fine-tuning to adapt to new situations or address limitations in the training data. Challenges arise from distributional shifts, causing bootstrap errors, as described in the literature on offline RL agent fine-tuning using conventional off-policy algorithms.

While model-free methods offer a simpler approach, they may struggle in complex environments and large state-action spaces. Researchers are exploring a spectrum of algorithmic innovations tailored for offline RL, ranging from adaptations of existing RL algorithms to novel training paradigms and the integration of cutting-edge machine learning and deep learning techniques. Significant strides have been made in both model-based and model-free offline RL. Model-based endeavors have concentrated on dynamics model learning for policy optimization, whereas model-free initiatives have focused on direct learning of policies and value functions from static datasets. Each approach grapples with distinct challenges, such as model bias in model-based methods and extrapolation errors in model-free techniques.

4 Data limitations

In offline RL, relying on pre-existing datasets poses significant challenges, particularly regarding data quality and diversity. The performance of offline RL algorithms depends critically on the dataset’s coverage and how well it represents the decision-making processes [8]. Datasets with limited diversity or poor decision-making strategies can significantly restrict the effectiveness of learned policies. Their static nature often introduces biases in policy learning, especially when the data comes from suboptimal or unknown behavior policies, making it more challenging to develop effective policies [20, 29]. A key challenge in the field is defining the conditions under which data coverage is sufficient to reliably learn optimal policies in an offline setting [30]. This issue is compounded by the absence of corrective feedback mechanisms inherent to online RL environments, which are pivotal for iterative policy refinement and adaptation to novel scenarios.

The objective of offline RL is to construct near-optimal policies from potentially flawed datasets, a stark contrast to the exploratory and interactive nature of online RL [31]. Developing robust policies in offline RL requires careful analysis of dataset characteristics and the deliberate application of learning algorithms. In this setting, the traditional balance between exploration and exploitation, central to online RL, must be reconsidered and adapted to the limitations of offline learning environments [20].

Bootstrapped Transformer (BooT) [32] addresses challenges in offline RL through sequence modeling, generating data with a Transformer model. Training is enhanced by bootstrapping with high-confidence sequence data. It explores autoregressive and teacher-forcing generation methods, alongside two bootstrapping strategies: one-time training on generated data and repetitive training with generated data. BooT algorithm is designed to integrate predictive dynamics and policies within the sequence generation model, eliminating the need for separate environment models. This approach is aimed at producing data that align with the offline dataset, particularly beneficial for proprioceptive observations like positions and velocities.

For training with the augmented dataset, BooT-o trains the model once with the generated data and discards them, whereas BooT-r continually integrates the generated data into training. High-quality trajectories for training are selected based on confidence scores, defined by the average log probability of the generated tokens. To stabilize the training, it applies a form of curriculum learning, starting with the original dataset and gradually incorporating self-generated trajectories, simplifying the process without intricate parameter tuning and showcasing effective results in their experiments.

5 Distributional shift: a historical challenge

Distributional shift in deep learning and RL has a longstanding history [33–35] and is not a recent research focus. In the context of deep learning, models are typically trained on large datasets with the assumption that future data will resemble the training set. However, when the model is deployed in the real world, it often encounters data that diverges from its training examples [36]. This can lead to decreased performance because the statistical properties of the deployment data differ from those the model learned during training. Distributional shift in RL involves changes between the distribution of states and actions the policy was trained on (offline data) and those it encounters in an online environment [37].

However, offline RL signifies a notable shift from traditional RL paradigms, necessitating innovative strategies to tackle its unique challenges [20], particularly regarding learning from static datasets and mitigating distributional shifts. Offline RL introduces distinct challenges such as covariate shift [38], distributional shift [37], and extrapolation error [39], among others. The endeavor to address these challenges is pivotal for the development of robust, efficient, and widely applicable learning algorithms within the realm of offline RL. The field is engaged in a concerted effort to overcome these challenges, with the aim of establishing robust, efficient, and generalizable learning algorithms [28, 40].

Despite its inherent challenges, offline RL holds significant transformative potential, with the capability to convert vast datasets into powerful decision-making frameworks across various sectors including healthcare [41], autonomous vehicles, and robotics [42, 43]. The constraints inherent in offline RL have spurred divergent research trajectories, such as the simulation-to-reality transfer techniques prevalent in robotics and autonomous driving. Yet, the reliance on high-fidelity simulators for such transfers accentuates the pressing need for the advancement of more resilient offline RL methodologies.

The issue of distributional shifts is exacerbated in the offline setting, where no new data is available to rectify extrapolation errors, leading to biases in Q-function estimation due to reliance on the visitation distribution of behavior policies. This necessitates careful consideration in balancing policy constraints [44] to ensure stability while maintaining flexibility for effective optimization. Furthermore, the lack of environmental feedback, crucial in online RL for iterative policy refinement, adds complexity to the learning process.

5.1 Distributional shift problem formulation in offline RL

Here, we explore the concept of distributional shift within the context of offline RL. The primary objective is to minimize the Bellman error, which arises from the action-value Bellman equation. The formalization is as follows:

$$J(\theta) = \mathbb{E}_{s,a,s' \sim D}\left[r(s,a) + \gamma \mathbb{E}_{a' \sim \pi_{\text{off}}(\cdot|s)}[Q^{\pi}_{\theta}(s',a')] - Q^{\pi}_{\theta}(s,a)\right] \tag{3}$$

where θ represents the parameters of learned Q-function. The objective $J(\theta)$ is only accurate when the behavior policy $\pi_\beta(a|s)$ equals the offline policy $\pi_{\text{off}}(a|s)$, because it ensures that the Q-function evaluates actions that were used during training. However, in practice, this equality is not feasible as the goal is to derive a new, improved policy π_{off} that surpasses the behavioral policy π_β, inevitably leading to distributional shifts. Additionally, even if π_{off} can accurately evaluate the training data objective, the induced state distribution might deviate due to compounding errors from sampling or function approximation, i.e., $d^{\pi_{\text{off}}}(s) \neq d^{\pi_\beta}(s)$. These issues are particularly problematic in offline RL, where continuous interaction to correct such deviations, as seen in off-policy RL, is not possible.

5.2 Out-of-distribution

OOD refers to scenarios where a machine learning model encounters data during inference that significantly deviates from the distribution of the training data. These deviations can occur in various forms, including covariate shift, prior probability shift, or concept shift. In essence, OOD data lies outside the manifold or statistical representation of the training dataset, making accurate prediction or classification inherently challenging [16, 17].

OOD detection involves identifying and appropriately handling these unseen data points to ensure robust model performance. In the context of generalized OOD detection, frameworks aim to classify OOD data based on varying characteristics such as novelty detection, anomaly detection, or open-set recognition. For instance, generalized OOD detection goes beyond traditional binary classifications of "in" vs. "out" distribution by addressing complex scenarios such as detecting subtle shifts in high-dimensional data (e.g., vision-language tasks) or managing cases with evolving distributions over time [14, 15].

The ability to generalize to OOD data is critical for real-world applications, where models often face unseen distributions in high-stakes domains. A growing body of research highlights the importance of designing algorithms capable of improving OOD robustness and developing evaluation metrics to quantify the extent to which models generalize to unseen distributions [15, 16].

6 Classification of existing methods

Research in offline RL presents a dynamic and evolving landscape, addressing unique challenges, applying innovative methods, and integrating theoretical insights with practical applications. This body of work reflects a field progressing toward fully realizing the potential of RL in offline settings, with significant implications for real-world problems. As shown in Fig. 2, we categorize the methodologies into two main groups: Q-restriction and policy-constraint approaches. Many studies within these frameworks also employ various uncertainty estimation techniques, which are examined in detail in the following subsections.

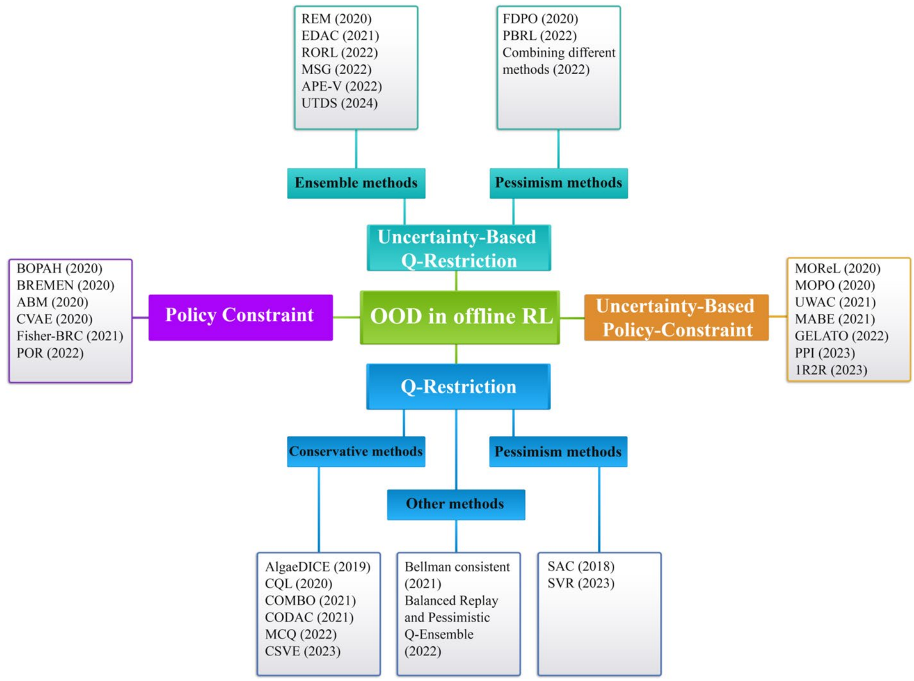
6.1 Q-Restriction approaches

Q-Restriction approaches involve modifying the Q-learning algorithm to be more conservative when estimating Q-values for actions that are not well-represented in the training data [12]. This is done to ensure that the learned policy does not overestimate the value of unseen actions, which can occur due to the distributional shift between the behavior and learned policies [45].

6.1.1 Conservative methods

In conservative Q-learning, the Q-values for OOD state-action pairs are penalized to ensure that the learned Q-function provides a lower bound on the true Q-function [46]. Conversely, value regularization modifies the value objective, introducing a penalty to the learned value function for more conservative estimates.

Fig. 2 The taxonomy of the proposed method within the context of OOD in offline RL



Kumar et al. [45] developed Conservative Q-learning (CQL), a method that learns a lower bound of the Q-function by adding value regularization terms, ensuring the state-value function underestimates values for all dataset states. This conservative estimate aims to mitigate the overestimation issue in Q-learning. CQL integrates a Q-value regularization mechanism into Bellman error optimization, balancing the minimization of Q-values for a predefined state-action distribution and the maximization for empirical data. This method applies to both Q-learning and actor-critic architectures [47, 48] and has proven effective in various tasks, including standard Gym environments and complex robotic control.

The core algorithmic procedure of CQL entails minimizing the expected Q-value over a designated distribution of state-action pairs, denoted by $\mu(s, a)$, which is aligned with the state marginal observed in the dataset. The iterative update of the Q-function is modulated by a tradeoff factor α, as delineated in the following formulation:

$$\hat{Q}^{k+1} \leftarrow \arg\min_{\hat{Q}} \alpha \mathbb{E}_{s\sim D, a\sim\mu(a|s)}[\hat{Q}(s,a)] + \frac{1}{2}\mathbb{E}_{s,a\sim D}\left[\left(\hat{Q}(s,a) - \hat{B}_\pi \hat{Q}^k(s,a)\right)^2\right] \tag{4}$$

where $\hat{B}_\pi$ is the empirical Bellman operator. To refine the estimation of $V_\pi(s)$, an additional term that maximizes Q-values under the behavior policy $\pi_\beta(a \mid s)$ is introduced, leading to an updated iterative process:

$$\begin{aligned}\hat{Q}^{k+1} \leftarrow \arg\min_{\hat{Q}} \alpha \left[\mathbb{E}_{s\sim D, a\sim\mu(a|s)}\left[\hat{Q}(s,a)\right] - \mathbb{E}_{s\sim D, a\sim\hat{\pi}_\beta(a|s)}\left[\hat{Q}(s,a)\right]\right] \\ + \frac{1}{2}\mathbb{E}_{s,a,s'\sim D}\left[\left(\hat{Q}(s,a) - \hat{B}_\pi \hat{Q}^k(s,a)\right)^2\right]\end{aligned} \tag{5}$$

CQL's approach emphasizes exploring unseen actions through a regularization term, providing a saddle point solution that can be unstable. To stabilize, Nachum et al. [49] suggest using entropy, resulting in a more stable training process. They introduced value regularization that depresses Q-values for actions from the learned policy, addressing overestimation in OOD actions. Despite its advantages, the method has limitations, including computational challenges for continuous actions requiring numerical integration, which is often addressed through Monte Carlo sampling. CQL generally outperforms policy constraint methods in complex tasks requiring the composition of sub-optimal behaviors. It's particularly effective in environments where learning to stitch together different tasks learned from prior data is essential.

Conservative Offline Model-Based Policy Optimization (COMBO), proposed by Yu et al. [50], enhances performance and generalizability in RL tasks by avoiding the complexities of explicit uncertainty quantification. COMBO focuses on generating conservative value function estimates for state-action pairs outside the dataset's support, ensuring robustness and safety by preventing overoptimistic value predictions. It penalizes Q-values from underrepresented state-action distributions while inflating those from more reliable pairs. This approach incorporates conservatism into policy optimization without direct uncertainty measures, streamlining the process and improving reliability.

Mildly Conservative Q-Learning (MCQ) by Lyu et al. [12] applies mild conservatism in offline RL, aiming to improve generalization without severely penalizing OOD actions. MCQ contrasts with existing methods that either restrict policies closely to the behavior policy or penalize value functions for OOD actions, often resulting in excessive pessimism. It actively trains OOD actions by assigning them proper pseudo Q values, ensuring that the learned policy performs at least as well as the behavior policy without incurring erroneous overestimation for OOD actions. Conditional Variational Auto-Encoder (CVAE) is been used in this method to estimate the behavior policy, facilitating the application of the MCQ operator. Policy optimization in MCQ endeavors to maximize the expected policy value, taking into account the minimum Q-value from the critic ensemble and imposing a penalty for high-entropy policies to foster exploration.

Unlike traditional methods that focus on conservative Q-function estimation, Chen et al. [51] presented Conservative State Value Estimation (CSVE), which shifts the focus from penalizing OOD actions in Q-function estimation to directly penalizing the V-function for OOD states. CSVE enhances state value estimation with conservative guarantees, facilitating improved policy optimization. CSVE employs Advantage Weighted Regression with state exploration for policy improvement, guiding the policy towards states with conservative value estimates and leveraging a dynamics model to sample OOD states for efficient value penalization and policy exploration.

Conservative Offline Distributional Actor-Critic (CODAC) [46] extends conservative Q-learning by penalizing OOD actions, using a penalty term to ensure conservative estimates of the quantiles of the learned return distribution, Z'_{k+1}. The conservative distribution evaluation is expressed as:

$$Z'_{k+1} = \arg\min_{Z} \alpha \cdot \mathbb{E}_{U(\tau),D(s,a)} \left[c_0(s,a) \cdot F^{-1}_{Z(s,a)}(\tau) \right] + \mathcal{L}_p(Z, \hat{T}_\pi Z'_k) \tag{6}$$

Here, $c_0(s,a)$ scales penalties for OOD actions, α adjusts penalty magnitude, $\mathcal{L}_p$ represents a loss function, and $F^{-1}_{Z(s,a)}$ denotes the inverse cumulative distribution function of Z for a state-action pair (s, a). CODAC is shown to converge to an estimate of the return distribution whose quantiles uniformly lower bound the true return distribution's quantiles. This property ensures that the algorithm is conservative across all potential returns, including the expected return and risk-sensitive objectives like CVaR. Practical Implementation CODAC is operationalized using deep neural networks to represent the quantile function, optimizing the objectives using distributional temporal-differences (TD) methods. The algorithm uses a quantile regression loss for optimizing the Wasserstein distance between the predicted and target quantile functions.

6.1.2 Pessimism methods

The concept of incorporating pessimism within value estimates has been explored as a means to address the distributional shifts between the training data and the target policy, as evidenced in several works [52, 53], highlighting the criticality of judicious data utilization and policy parameterization. Pessimistic methods, which rely on pessimistic extrapolation from offline data, aim to maximize the rewards the trained agent would receive in the worst possible MDP consistent with the observed dataset [54].

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Lee et al. [55] introduced a balanced replay strategy for offline RL that prioritizes online samples while incorporating near-on-policy samples from offline datasets. This method adjusts the sampling strategy to favor experiences closer to those encountered in online interactions, enhancing policy evaluation and value estimation in novel situations. The approach employs a pessimistically trained Q-function ensemble to maintain a conservative outlook on value estimations, particularly for unfamiliar state-action pairs, thus stabilizing the policy fine-tuning process against over-optimism. The balanced replay technique efficiently integrates online and offline learning phases, addressing distributional shifts by weighting samples based on their relevance to the current policy. This is complemented by an ensemble approach that averages multiple Q-function estimates to improve generalization and reduce error impacts.

Xie et al. [54] proposed the Bellman-consistent pessimism method, which encodes a systematic form of pessimism compatible with any function approximation and MDP. This method avoids the excessive conservatism of traditional bonus-based pessimism, which can hinder effective policy discovery. Instead of using point-wise lower bounds, it implements pessimism at the initial state based on functions consistent with Bellman equations. They introduced an efficient information-theoretic algorithm that leverages offline data to compute lower bounds on policy values and automatically adapts to the optimal bias-variance trade-off. Additionally, they developed a practical algorithm called Pessimistic Soft Policy Iteration (PSPI), which is computationally efficient and theoretically robust, using regularization and mirror descent updates for policy optimization. This algorithm uses regularization and mirror descent updates for efficient policy optimization. Here is the empirical estimate of Bellman error where $\mathcal{L}$ is a loss function measuring the discrepancy between f' and f under policy π over dataset D.

$$\mathcal{E}(f,\pi;D) = \mathcal{L}(f,f,\pi;D) - \min_{f'\in F} \mathcal{L}(f',f,\pi;D) \tag{7}$$

6.1.3 Other methods

For policy regularization, Haarnoja et al. [47] proposed entropy regularization to control the stochasticity of the optimal policy, enhancing training robustness by preventing premature convergence.

Mao et al. [56] introduced Supported Value Regularization (SVR) for offline RL, which distinguishes in-distribution (ID) from OOD actions using an importance sampling bias and penalty mechanism. SVR penalizes Q-values for OOD actions while maintaining standard Bellman updates for ID actions, balancing conservatism with optimal policy convergence. It automatically applies penalties based on importance sampling, avoiding manual discrimination between ID and OOD actions. The policy evaluation operator in SVR, designed as a contraction mapping in tabular MDP environments, ensures unbiased Q-value evaluation for ID actions and conservative underestimation for OOD actions, promoting strict policy enhancement. The SVR loss function incorporates a regularization term to differentiate the treatment of ID and OOD actions, enhancing its effectiveness in offline RL.

Table 3 provides a comprehensive summary of the reviewed methods within the Q-restriction category.

6.2 Uncertainty-based Q-restriction approaches

In offline RL, uncertainty estimation plays a pivotal role in enhancing policy robustness, particularly against OOD actions and distributional shifts. Techniques such as bootstrapped Q-networks [73], Bayesian approaches, and ensemble methods are employed to quantify uncertainty, aiding in addressing challenges like extrapolation errors. These methods focus on providing more reliable value function predictions, essential for decision-making in scenarios where the dataset may not fully represent the state-action space. Despite their potential, these approaches face difficulties in offering precise uncertainty estimates, especially when using complex function approximators like deep neural networks.

The research community is actively exploring advanced uncertainty quantification and regularization strategies to refine policy and value function updates. If we can estimate the epistemic uncertainty of the Q-function,

Table 3 Summary of Q-restriction methods

Refs.	Year	Method	Dataset	Evaluation metrics	Compared results to	Code	Ablation
[47]	2018	SAC	HalfCheetah, Hopper, Walker2d, Ant, Humanoid	Average Return, Sample Efficiency, Stability Across Random Seeds	TD3 [48], DDPG [27], PPO [57], SQL [58], Duan et al. [59]	[60]	✓
[49]	2019	Algae DICE	Four Rooms domain, MuJoCo, OpenAI Gym	Average Return, Sample Efficiency, Stability Across Random Seeds	SAC, TD3, DDPG	–	–
[45]	2020	CQL	AntMaze, Adroit, kitchen, Atari games	Average Return, Sample Efficiency, Stability Across Random Seeds	BEAR [53], BRAC [61], SAC, REM, QR-DQN [62]	–	✓
[50]	2021	COMBO	halfcheetah-jump, ant-angle, walker-walk, sawyer-door	Average Return, Stability Across Random Seeds, Model Error	BEAR, BRAC, SAC, CQL, MOPO, MOReL, LOMPO [63]	–	–
[54]	2021	Bellman-consistent pessimism	–	Computational complexity (O)	Liu et al. [64]	–	–
[46]	2021	CODAC	HalfCheetah, Hopper, Walker2d	Average Return, Sample Efficiency, Stability Across Random Seeds	CQL	[65]	✓
[55]	2022	Balanced Replay and Pessimistic Q-Ensemble	halfcheetah, hopper, walker2d	Average Return, Stability Across Random Seeds, AUROC	AWAC, BCQ [52], SAC	[66]	✓
[12]	2022	MCQ	MuJoCo	Average Return, Sample Efficiency, Stability Across Random Seeds	BC [67], SAC, CQL, AWAC, UWAC, TD3+BC, IQL [68]	[69]	–
[56]	2023	SVR	HalfCheetah, Hopper, Walker2d	Normalized Return, Stability Across Random Seeds	BC, BCQ, BEAR, OneStep RL [70], TD3+BC, CQL, AWAC, UWAC, IQL	[71]	✓
[51]	2023	CSVE	Gym, Adroit, Half-Cheetah, Hopper, Walker2d	Average Return, Sample Efficiency, Stability Across Random Seeds	CQL, COMBO, AWAC, UWAC, BEAR, TD3+BC, IQL	[72]	✓

we expect this uncertainty to be substantially larger for OOD actions, and can therefore utilize these uncertainty estimates to produce conservative target values in such cases. Learning an uncertainty set or distribution over possible Q-functions from the dataset enhances this approach.

6.2.1 Ensemble methods

Uncertainty in Q-function estimates can be addressed through an ensemble of Q-functions. Agarwal et al. [74] proposed a naive approach which uses disjoint dataset partitions to train multiple Q-functions, approximating the Q-function distribution through sample means and variances. It also proposes the Random Ensemble Mixture (REM) method, using a random convex combination of Q-functions to estimate Q-values, which shows effectiveness in datasets with high coverage but may underestimate uncertainty, potentially leading to sub-optimal OOD action selection. Diverging from Ensemble-DQN's average estimation, REM uniquely optimizes Q-value combinations to minimize TD errors, providing a fresh perspective on Q-value estimation and algorithmic progress in offline RL.

Ensemble-Diversified Actor Critic (EDAC) [75] addresses the issue of error propagation from OOD actions by penalizing uncertainty, different from typical algorithms that enforce policy or Q-function regularization towards the dataset. It employs an ensemble of Q-function networks to quantify the epistemic uncertainty in Q-value estimates without relying on data distribution estimation or sampling. By implementing clipped Q-learning which selects the minimum value as the target, mitigates overestimation errors and expanding the number of Q-ensembles, it penalizes high-variance state-action pairs, favoring actions within the dataset and imposes greater penalties on OOD actions.

Robust Offline RL (RORL) [76], enhanced policy and value function robustness in offline RL by employing conservative smoothing techniques. RORL addresses the challenge of overestimation errors, particularly for

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OOD states, through pessimistic bootstrapping. It applies smooth regularization to both policy and value functions, using perturbation methods on state-action pairs to enforce Q-function smoothness and minimize variance in Q-estimates. The methodology includes a Robust Q-function formulation, utilizing an ensemble of Q-functions to establish learning targets that balance conservatism and exploration. RORL's approach integrates uncertainty quantification and differential loss functions to achieve conservative value function estimates and enhance policy stability, marking a significant advancement in offline RL methodologies.

There was a significant issue in ensemble-based RL algorithms where shared pessimistic target values may inadvertently lead to optimistic value estimates. To rectify this, Model Standard-deviation Gradients (MSG) algorithm [77] assigns unique, independently computed targets to each ensemble member, thus ensuring more reliable and genuinely conservative value estimates. MSG emphasizes individual training for each Q-function within the ensemble, leveraging the lower confidence bound (Q-LCB) to guide policy optimization. It avoids the pitfalls of shared targets by allowing each ensemble member to contribute distinctively to the overall policy evaluation, factoring in the inherent uncertainty in value function estimation. MSG algorithm's emphasis on independent targets and the use of Q-LCB for policy optimization marks a significant advancement in addressing ensemble-based RL algorithms' challenges, ensuring more accurate and conservative policy evaluations.

Adaptive Policies with Ensembles of Value Function (APE-V) [78] introduced a Bayesian framework that underscores the need for adaptive policies that adjust based on accumulated episode information. They propose APE-V, a practical, model-free algorithm designed to approximate optimal adaptive policies within this framework. APE-V employs an ensemble of value functions to quantify the agent's uncertainty and guides policy adaptation accordingly. The core objective is to maximize expected returns across a posterior distribution of MDPs, reflecting the uncertainties of the offline dataset. It emphasizes the significance of adaptability and uncertainty estimation in enhancing offline RL policy effectiveness.

Multi-Task Data Sharing (MTDS) [79] proposed Uncertainty-based Data Sharing (UTDS), a novel method in offline RL that integrates diverse datasets from multiple tasks into a unified mixed dataset without selective curation. UTDS employs ensemble-based uncertainty quantification to generalize across single-task and multi-task RL scenarios by combining datasets and using an ensemble of Q-functions to assess uncertainty. It innovatively applies uncertainty quantification for pessimistic value iteration, penalizing value updates based on uncertainty to mitigate policy deviations and ensure robust learned policies. UTDS extends this approach to penalize Q-functions for RL actions, enhancing its effectiveness in handling unfamiliar data regions. Overall, UTDS enhances offline RL performance by facilitating MTDS, offering a scalable approach to improve RL algorithms' reliability and applicability across diverse tasks. Charpentier et al. [80] proved that disentangling aleatoric and epistemic uncertainty is important in RL tasks and can improve generalization and performance. They combined different types of uncertainty estimation methods such as MC dropout, ensemble, deep kernel learning, and evidential networks [81] and proposes a practical methodology to evaluate uncertainty in model-free RL based on detection of OOD environments and generalization to perturbed environments. An approach to estimating uncertainty involves considering the distribution of Q-functions over a dataset D, denoted as $P_D(Q^\pi)$. We can adjust our policy gradient objective to penalize the uncertainty represented by $P_D(Q^\pi)$ as follows:

$$J(\theta) = \mathbb{E}_{s,a\sim D}\left[\mathbb{E}_{Q^\pi\sim\mathcal{P}_D(\cdot)}\left[Q^\pi(s,a)\right] - \alpha U_{\mathcal{P}_D}(\mathcal{P}_D(\cdot))\right] \tag{8}$$

where $U_{P_D}(\cdot)$ is an uncertainty measure for P_D.

6.2.2 Pessimism methods

Buckman et al. [82] introduced Fixed-Dataset Policy Optimization (FDPO), a theoretical framework for offline RL that addresses the overestimation errors in naïve FDPO strategies. These strategies assume accurate environment modeling from finite datasets, which often isn't realistic. The researchers highlighted the "Pessimism Principle," which advocates selecting policies that perform well under worst-case scenarios to avoid overestimation.

This principle results in underestimation errors but promotes robust policy optimization. Proximal pessimistic algorithms implement this by penalizing policies that deviate from the empirical policy observed in the data. Additionally, uncertainty-aware pessimistic algorithms incorporate state-wise Bellman uncertainty into policy penalization, fostering cautious value estimation. This is achieved through a penalization mechanism that adjusts the naïve fixed-point function by incorporating Bellman uncertainty, thereby instilling a layer of pessimism in value estimations.

$$f_{ua}(v^{\pi}) = A^{\pi}\left(r_D + \gamma P_D v^{\pi}\right) - \alpha u^{\pi}_{D,\delta} \tag{9}$$

where α is the pessimism hyper parameter and $u^{\pi}_{D,\delta}$ is the Bellman uncertainty function. Conversely, proximal pessimistic algorithms impose penalties based on the deviation from empirically observed policies within the dataset, thereby anchoring policy selection within the bounds of empirical data and reducing the propensity for overestimation errors. This framework not only exposes the limitations of current FDPO methods but also guides the development of more reliable offline RL approaches. By highlighting the contrast between uncertainty-aware and proximal approaches, the authors emphasizes the potential for enhanced generalizability and efficacy through the adoption of non-trivial uncertainty functions, a prospect currently constrained by the computational limitations of neural networks in capturing complex uncertainties.

Pessimistic Bootstrapping for offline RL (PBRL), as introduced by Bai et al. [83], innovatively integrates uncertainty quantification into Q-function updates to enhance the robustness and reliability of RL algorithms in offline settings. PBRL utilizes a Q-function ensemble to estimate epistemic uncertainty, leveraging variance among these functions to penalize OOD actions effectively. Unlike traditional conservative methods in offline RL, PBRL employs uncertainty quantification directly for penalization, distinguishing its approach. PBRL also introduces an OOD sampling technique to directly regulate OOD behavior, bolstering the stability of the ensemble Q-functions by incorporating additional OOD data points during training. This method ensures that the critic ensemble adjusts target values based on uncertainty levels, thereby promoting reliable policy development in diverse offline RL applications.

Table 4 Summary of uncertainty-based Q-restriction methods

Refs.	Year	Method	Dataset	Evaluation metrics	Compared results to	Code	Ablation
[74]	2020	REM	Creating a DQN Replay Dataset using Atari 2600 Games	Normalized scores, Average Return	QR-DQN, Averaged Ensemble-DQN	[84]	✓
[82]	2020	FDPO	2000 and 500000 transitions	Performance score	–	[85]	–
[75]	2021	EDAC	HalfCheetah, Hopper, Walker2d	Average Return, Runtime, GPU memory consumption	BC, SAC, CQL, REM	[86]	–
[80]	2022	Combining different types of uncertainty estimation methods	CartPole, Acrobot, LunarLander, Gym environments	Training time, Testing time, Testing reward	–	–	–
[83]	2022	PBRL	HalfCheetah, Hopper, Walker2d	Normalized scores, Stability Across Random Seeds	BEAR, TD3+BC, CQL, UWAC, MOPO	[87]	✓
[76]	2022	RORL	HalfCheetah, Hopper, Walker2d	Average Return, Stability Across Random Seeds	BC, SAC, CQL, PBRL, EDAC	–	✓
[77]	2022	MSG	HalfCheetah, Hopper, Walker2d, Ant agent, RL Unplugged	Average Return, Stability Across Random Seeds	BC, CRR [88], MuZero Unplugged [26]	–	✓
[78]	2022	APE-V	HalfCheetah, Hopper, Walker2d, Locked Doors, Procgen Mazes	Average Return, Stability Across Random Seeds, Success rates	BC, SAC, CQL, REM, IQL	–	✓
[79]	2024	UTDS	Providing a suite of domains built on DeepMind Control Suite and collecting multi-task datasets	Runtime, GPU memory, Number of parameters, Average Return, Stability Across Random Seeds	CQL, CDS [89]	[90]	–

Table 4 presents a concise overview of uncertainty-based Q-restriction methods.

6.3 Policy constraint approaches

Addressing extrapolation errors from OOD actions is pivotal in offline RL, notably the use of model-based RL [91, 92] techniques and enhancements to off-policy RL algorithms. A common strategy is policy regularization, which restricts the policy to actions present in the training dataset to mitigate the evaluation of OOD actions.

However, this may limit the policy's ability to generalize and discover optimal strategies, constrained by the accuracy of behavior policy estimation. A significant portion of offline RL research aims to closely align the learned policy with the behavior policy through techniques like Kullback–Leibler (KL) divergence, ensuring policy learning stability in the absence of new data. These approaches necessitate careful selection of hyper-parameters and regularization terms, with Wu et al. [61] emphasizing their critical role in the success of offline RL regularization strategies. A significant challenge remains that constraints within the data distribution do not fully mitigate the errors in Q-value estimation caused by successive off-policy evaluations [70].

The performance of RL algorithms is significantly influenced by the accuracy with which the behavior policy is estimated. This task becomes particularly challenging in real-world scenarios characterized by highly multi-modal behaviors. To mitigate these challenges, methods that use only samples without explicit behavior policy estimation, offer a promising approach. Additionally, function approximation issues can inhibit the Q-function from learning meaningful values, especially when OOD actions are included in target values. This problem is exacerbated when the dataset is small, leading to cumulative errors, or when the dataset's distribution is too narrow, potentially yielding brittle, non-generalizable solutions.

An ongoing crucial question in the field is how to effectively adjust the level of conservatism to balance the risks of overestimation with the potential pitfalls of avoiding unfamiliar actions, ensuring the algorithm remains both effective and robust. Some of the methods use importance sampling which can suffer from high variance, which can dramatically increase in a sequential setting due to the multiplication effect, leading to exponential blowup. To mitigate this, some strategies avoid the multiplication of weights, but challenges remain.

Batch Optimization of Policy And Hyper-parameter (BOPAH) [93] utilizes gradient-based hyper-parameter optimization on a validation dataset to improve batch RL [94] performance. This method leverages a state-dependent KL-regularized RL framework to minimize the divergence between the estimation and behavior policies, effectively mitigating the covariate shift.

Behavior-Regularized Model-Ensemble (BREMEN) [95] tackles model bias and distribution shift in offline RL by combining ensemble dynamics models with behavior cloning (BC) and trust-region policy updates. This method merges Dyna-style model-based RL with behavior regularization to train policies that adhere closely to observed behaviors in the training data. By generating simulated roll-outs using an ensemble of models and initializing policies through BC, BREMEN ensures policy refinement through trust-region updates. The framework employs KL-based trust-region optimization to balance expected returns with policy consistency, ensuring minimal divergence between consecutive policies and maintaining fidelity to data-collection policy.

The Advantage-weighted Behavior Model (ABM) [96] constrains the policy optimization to the state-action space covered by the batch data, enhancing the stability and reliability of off-policy learning from static datasets. ABM serves as a learned prior, guiding the RL policy towards actions with a historical precedent of success, thereby improving task-specific performance. The core methodology involves a policy iteration process where the policy enhancement phase is deliberately restricted to maintain proximity to the empirical behavior manifested within the batch data. This represents a significant departure from conventional Q-learning paradigms in batch RL, introducing a methodological innovation aimed at stabilizing the policy refinement phase. ABM's integration as a behavioral prior within the policy iteration cycle is a novel contribution, ensuring a balanced exploration-exploitation trade-off and fostering the utilization of diverse or imperfect batch data for effective RL. The methodology is formalized through the following equations, where D_μ is the behavior data and θ_{bm}

is optimized to maximize the likelihood of observed actions within the dataset, essentially learning a behavior model that reflects the data:

$$\theta_{bm} = \arg\max_{\theta_{bm}} \mathbb{E}_{\tau \sim D_\mu} \left[\sum_{t=1}^{|\tau|} \log \pi_{\theta_{bm}}(a_t \mid s_t) \right] \tag{10}$$

Furthermore, θ_{abm} is optimized to bias the RL policy towards data-supported and task-beneficial actions, employing a weighted behavior model to refine policy decisions:

$$\theta_{abm} = \arg\max_{\theta_{abm}} \mathbb{E}_{\tau \sim D_\mu} \left[\sum_{t=1}^{|\tau|} \log \pi_{\theta_{abm}}(a_t \mid s_t) \cdot f\left(R(\tau_{t:N}) - \hat{V}^{\pi_i}(s) \right) \right] \tag{11}$$

On the other hand, Rezaeifar et al. [97] proposed method subtracts a prediction-based exploration penalty instead of adding a conventional exploration bonus, discouraging the selection of uncertain actions to maintain alignment with the dataset's support. Utilizing a Variational Auto-Encoder (VAE) trained on offline data, this approach which emphasizes behavior regularization, leverages reconstruction error as a penalty, aiding in distinguishing between ID and OOD actions.

Fisher-BRC (Behavior Regularized Critic) [98] emphasizes regularizing the critic component of actor-critic algorithms. The critic is parameterized as the log-behavior-policy (which generated the offline data) plus a state-action value offset term. This offset term is key to their approach. They suggest using a gradient penalty as a regularizer for the offset term and demonstrate its equivalence to Fisher divergence regularization, drawing connections to score matching and energy-based model literature. By drawing parallels between Fisher-BRC and CQL, the research delineates how CQL's KL divergence regularization between a Boltzmann policy and the behavior policy can be conceptually aligned with Fisher-BRC's regularization approach. This insight underscores the broader applicability of behavior regularization in offline RL and highlights Fisher-BRC's computational efficiency, particularly in circumventing the intensive numerical integration often required by CQL.

Policy-guided Offline RL (POR) [99] proposed decomposing the conventional RL task into two sub-tasks handled by the guide-policy and execute-policy which simplifies the learning problem for each policy and enhances training stability and combines the stability of imitation-style methods with the potential for OOD generalization. The guide-policy learns the optimal next state for maximizing rewards, while the execute-policy learns to execute actions that lead to desired next states. This separation allows for leveraging state-compositionality from the dataset for logical OOD generalization. Unlike prior imitation-based methods that focus on action compositionality, POR enables state-compositionality, allowing the model to logically generalize beyond the actions present in the dataset and potentially surpass the dataset's limitations. Value function learning for guide-policy equation represents the asymmetric loss used for training the state value function V(s) for the guide-policy. Also, learning objective for guide-policy maximizes the expected value of the next state as suggested by the guide-policy g_ω while maintaining similarity to an estimated behavior guide-policy g_μ, controlled by the trade-off parameter α.

$$\max_{\omega} \mathbb{E}_{s \sim D, s' \sim g_\omega(s)} \left[V_\phi(s') + \alpha \log g_\mu(s'|s) \right] \tag{12}$$

The details of policy constraint methods are presented in Table 5.

6.4 Uncertainty-based policy-constraint approaches

Model-Based Offline RL (MOReL) [92] introduced the Pessimistic Markov Decision Process (P-MDP) to segment the state space into known and unknown regions, crucial for creating a conservative learning environment. This approach aims to mitigate model exploitation risks by penalizing exploration into uncertain territories, addressing challenges in offline RL where model imperfections can lead to unreliable policy decisions. MOReL distinguishes well-represented from underrepresented state-action pairs within the dataset using uncertainty-aware

Table 5 Summary of policy constraint methods

Refs.	Year	Method	Dataset	Evaluation metrics	Compared results to	Code	Ablation
[93]	2020	BOPAH	MuJoCo	Mean (normalized) performance, The conditional value at risk performance (CVaR)	BasicRL, RaMDP [100], Robust MDP [101, 102], SPIBB [103]	–	–
[95]	2020	BREMEN	Ant, HalfCheetah, Hopper, Walker2d	Average cumulative reward	BC, BCQ, BRAC, SAC, ME-TRPO [104]	[105]	–
[96]	2020	ABM	Cheetah, Hopper, Quadruped	Average Return, Stability Across Random Seeds	BCQ, BEAR	–	✓
[92]	2020	CVAE	HalfCheetah, Hopper, Walker2d, Human dataset	Average Return, Stability Across Random Seeds	BCQ, BEAR, BRAC, CQL, TD3+BC	–	–
[98]	2021	Fisher-BRC	Cheetah, Hopper, Quadruped	Average Return, Stability Across Random Seeds	BC, BRAC, CQL, MBOP [106]	[107]	✓
[97]	2022	POR	AntMaze, Cheetah, Hopper, Quadruped	Averaged normalized scores, Stability Across Random Seeds	BCQ, BEAR, BRAC, AWR [108], CQL	[109]	–

state discretization. It then optimizes a near-optimal policy within this pessimistically bounded environment, ensuring cautious reward estimation to prevent overoptimistic assessments in unfamiliar regions. By integrating pessimism into decision-making, MOReL enhances policy robustness, utilizing BC for initial policy approximation and subsequent planning within the P-MDP framework to effectively utilize pre-existing datasets for decision-making in complex environments.

Model-based RL with Adaptive Behavioral Priors (MABE) [110] merges the paradigms of model-based RL with adaptive behavioral priors. This approach deviates from traditional reliance on uncertainty quantification by focusing on the synthesis of neural network-based dynamics models and behavioral priors, which are effectively policies approximating actions within the offline dataset to enhance policy generalization both within and across domains. Central to MABE is its unique methodology that encompasses policy optimization through model-based RL, regularization via adaptive behavioral priors, and an optional component of uncertainty quantification for detailed policy refinement. The framework's standout feature is its capacity for cross-domain behavioral transfer, facilitated by the strategic application of learned dynamics models alongside the transferability of behavioral priors across diverse datasets.

Model-based Offline Policy Optimization (MOPO) algorithm [91], a novel adaptation of model-based RL techniques, distinguished by its integration of uncertainty-penalized rewards derived from the dynamics model. MOPO is designed to maximize a lower bound on the expected return of a policy under the true dynamics of a MDP, while carefully navigating the balance between exploiting the available batch data and the inherent risks of venturing into regions beyond the data's support. The algorithm employs neural network-based dynamics models that output Gaussian predictions, utilizing the predicted variance as a mechanism for uncertainty estimation. This uncertainty serves as a pivotal component in MOPO's strategy, acting as a penalty factor in the reward function. This penalization framework enables MOPO to cautiously extend its policy's exploration to OOD states and actions, mitigating potential risks associated with model inaccuracies. MOPO's methodological framework includes several key steps: quantifying the uncertainty present in the dynamics model, adjusting the reward function with a penalty reflective of this uncertainty to yield a conservative return estimate, and employing standard RL techniques to optimize this adjusted return estimate. This approach effectively ensures that the algorithm maintains a prudent exploration strategy, emphasizing safety and robustness in decision-making.

$$\eta_M(\pi) \geq \bar{\mathbb{E}}_{(s,a)\sim\rho^{\pi}_{\hat{T}}[r(s,a)-\gamma|G_{\pi M_c}(s,a)|]} \tag{13}$$

The mathematical foundation of MOPO is encapsulated in its formulation of an uncertainty-penalized reward function, denoted as $\tilde{r}(s,a)$, which underpins a conservative estimate of the policy's return. This formulation is crucial for achieving a balance between exploration and exploitation, providing a rigorous basis for the algorithm's operation within the confines of offline data. MOPO's innovative incorporation of model uncertainty into the reward mechanism marks a significant advancement in the domain of offline RL, enhancing the capacity for generalization beyond the limitations of traditional model-free approaches.

Building on the MOPO [91] framework, Tennenholtz and Mannor [111] introduced a method that penalizes rewards in OOD areas by estimating model errors, thus forming a penalized MDP. This is a hybrid approach for uncertainty estimation in offline RL by merging parametric methods like bootstrap ensembles [73] with non-parametric techniques such as k-nearest neighbors (k-NN). This strategy is designed to enhance model-based offline RL by quantifying uncertainty, particularly useful in data-sparse regions. Riemannian metric which is introduced based on an encoder-decoder model, to quantify both epistemic and aleatoric uncertainties evaluates the geodesic distance to k-NN in the dataset, aiding in assessing the reliability of OOD samples. The method employs k-NN to approximate model errors as a measure of uncertainty and leverages a generative latent model with an encoder and a decoder to further refine uncertainty estimation.

Uncertainty Weighted Actor-Critic (UWAC) [112] enhance offline RL by identifying and attenuating the influence of OOD state-action pairs. UWAC employs a dropout-based uncertainty estimation mechanism within the offline RL context, offering a pragmatic solution for integrating uncertainty estimation with minimal computational overhead. This method innovatively adjusts Bellman updates in actor-critic algorithms to de-emphasize contributions from state-action pairs with high uncertainty levels. Central to UWAC are the application of Monte Carlo dropout for uncertainty quantification and the employment of an uncertainty-weighted policy distribution, providing a Bayesian perspective to the Q-function optimization by maximizing the posterior distribution Ensemble methods play a crucial role in RL, particularly in mitigating overestimation biases in Q-functions, fostering exploration, and minimizing bootstrap error propagation. Adapted for both model-free and model-based RL frameworks, these methods are instrumental in tackling the distinct challenges presented by offline RL.

1R2R algorithm [113] addressed both epistemic uncertainty, related to data scarcity, and aleatoric uncertainty, inherent in environmental randomness, in offline RL. It uses ensemble models to represent uncertainty and penalizes actions with variable outcomes, discouraging risky decisions. The goal is to avoid distributional shifts and enhance risk-averse behavior, all within a bayesian MDP framework where the posterior distribution over MDPs is approximated from an offline dataset. The core mechanism is adversarially modifying the transition distribution during model roll-outs to induce risk aversion. The algorithm utilizes an ensemble of neural networks for transition and reward approximation, with policy training via SAC.

To address both immediate and future trajectory uncertainties in offline RL, a novel approach called Pessimistic Policy Iteration (PPI) [114] was proposed. It incorporates a global uncertainty estimator that adjusts policies or Q-values based on assessed uncertainty, promoting a more conservative strategy in high-uncertainty regions. It differentiates between policy-constraint [61] and Q-restriction [115] techniques by using global uncertainty to modulate policies for cautious exploration or to refine penalties for OOD actions, respectively. The global uncertainty is dynamically updated, taking into account both local uncertainties and the anticipated uncertainties of future state-action pairs. This approach enables a more granular and flexible handling of uncertainties, enhancing policy reliability and adaptation in uncertain environments. It is in contrast with IQL [68] that avoids evaluating actions outside the dataset, yet allows policy improvement through generalization by iteratively fitting the upper expectile value function and integrating it into a Q-function. Policy improvement with Q-value penalization with uncertainty penalizes the Q-values based on the level of uncertainty, thereby encouraging conservative Q-value estimations in regions of high uncertainty.

Table 6 provides a summary of methods classified as uncertainty-based policy-constraint approaches.–

7 Offline RL benchmarks

Previous work in offline RL often used online RL algorithms to train the behavior policy, utilizing data from replay buffers or rollouts from the final policy to create a static dataset. However, such data might originate from non-Markovian policies like human-driven or handcrafted strategies, making these datasets potentially

Table 6 Uncertainty-based policy-constraint methods

Refs.	Year	Method	Dataset	Evaluation metrics	Compared results to	Code	Ablation
[92]	2020	MOReL	HalfCheetah, Hopper, Walker2d, Ant	Average Return, Stability Across Random Seeds	BCQ, BEAR, BRAC	[116]	–
[91]	2020	MOPO	HalfCheetah, Hopper, Walker2d, AntMaze, Adroit, Kitchen	Average Return, Stability Across Random Seeds	BC, BEAR, BRAC, SAC	–	✓
[112]	2021	UWAC	HalfCheetah, Hopper, Walker2d, Adroit	Average Return, Stability Across Random Seeds	BC, MOPO, MOReL, BEAR, AWR, BRAC, BCQ, CQL, REM, AlgaeDICE [49]	–	✓
[110]	2021	MABE	HalfCheetah, Hopper, Walker2d	Average Return, Stability Across Random Seeds	BC, MOPO, MOReL, BRAC, CQL, SAC	–	✓
[111]	2022	GELATO	HalfCheetah, Hopper, Walker2d, Autonomous vehicle environments highway-env	Average Return, Stability Across Random Seeds	MOPO, SAC, MBPO [117]	–	–
[114]	2023	PPI	HalfCheetah, Hopper, Walker2d	Average Return, Stability Across Random Seeds, Robustness	TD3+BC, CQL	–	–
[113]	2023	1R2R	HalfCheetah, Hopper, Walker2d, Currency Exchange, HIV Treatment	Average Return, Stability Across Random Seeds	ORAAC [118], CODAC, RAMBO [28], COMBO, MOPO, TD3+BC, IQL, CQL, ATAC [115]	[119]	–

unrepresentative of real-world scenarios where offline RL could be applied. In an ideal situation, real-world datasets would be used to evaluate offline RL algorithms.

Narrow and Biased Data Distributions manage data from human or crafted policies to prevent divergence into OOD states. Undirected and Multitask Data challenge algorithms to synthesize segments into viable solutions, testing non-multistep dynamic programming methods. Sparse Rewards pose credit assignment challenges, necessitating simple, effective task completion criteria. Suboptimal Data pushes algorithms beyond imitation, enhancing their refinement capabilities. Nonrepresentable Behavior Policies arise when behavioral complexity exceeds the model's capacity, requiring adaptable network architectures. Non-Markovian Behavior Policies, typical in human-derived systems, challenge RL's Markovian assumptions. Realistic Domains call for high-fidelity simulations to ensure algorithm effectiveness, circumventing the impracticality of real-world testing. Nonstationarity demands dynamic adaptation to environmental changes like sensor malfunctions or mechanical wear, ensuring robustness in diverse, dynamic scenarios.

Desired properties for offline datasets to advance offline RL in realistic applications, based on insights from Fu et al. [120] and Gulcehre et al. [121] are handling narrow and biased data distributions, undirected and multitask data, sparse rewards, sub-optimal data, non-representable behavior policies, non-Markovian behavior policies, realistic domains, and non-stationarity. Each attribute addresses specific challenges in offline RL, such as ensuring algorithms do not diverge when faced with biased data or enabling them to stitch together data fragments to solve tasks that individual trajectories do not solve. Key environmental characteristics sought in offline RL datasets are continuous action and state spaces, stochastic dynamics, and partial observability. Continuous spaces, challenging due to their inability to visit every state, require agents to generalize beyond observed states and actions. Stochastic dynamics mimic the natural world's inherent randomness, often stemming from model limitations. Partial observability, common when full domain knowledge is unavailable, is crucial for applying offline RL in realistic scenarios like POMDPs.

The D4RL and RL Unplugged benchmarks, covering a range of tasks from Gym-MuJoCo, mazes, and dexterous manipulation to robotic tasks and autonomous driving (CARLA [122]) offer environments and datasets that cover many of the qualities essential for an effective offline RL dataset. Almost all of the algorithms discussed in this survey have used D4RL benchmark specially the GYm-MuJoCo environment and some used AntMaze and Adroit [123] environments aswell. RL Unplugged primarily features datasets from behavior policies trained online, contrasting with D4RL's inclusion of diverse policies like non-Markovian ones. RL Unplugged's data typically solves tasks directly, reducing the need for algorithms to perform task stitching. Evaluation-wise, D4RL does not specify protocols, whereas RL Unplugged uses both online and offline validations, with the former allowing for online samples which deviates from the typical offline RL setup. D4RL also offers datasets from a range of policy types (random to expert) enabling testing across different data qualities, whereas RL Unplugged generally provides data from more successful policies, limiting behavior policy variety.

7.1 D4RL benchmark

D4RL (Datasets for Deep Data-Driven Reinforcement Learning) offers a comprehensive collection of benchmark datasets tailored for advancing DRL research. These datasets span diverse environments including robotics (e.g., manipulation tasks), control scenarios (e.g., balancing, navigating), and game-playing domains (e.g., Atari, Go). Each dataset comprises extensive trajectories generated by executing policies within these environments, serving as training data for RL algorithms aimed at policy improvement. The primary objectives of D4RL is to establish a standardized benchmark for evaluating data-driven RL algorithms, foster the development of novel algorithms capable of learning from large-scale experience data, and enable comparative analyses across different RL approaches on common tasks. Key features include large-scale and diverse datasets, high-quality data generated from both simulation and real-world sources, standardized formats for accessibility, and predefined evaluation metrics such as return and success rate [120]. This benchmark is available on GitHub [124]. Some key aspects of D4RL are presented in Table 7.

7.1.1 OpenAI Gym

OpenAI Gym [125] is a toolkit developed by OpenAI for developing and comparing RL algorithms. It provides a collection of standardized environments and benchmarks to facilitate RL research. Here are some key aspects of OpenAI Gym:

- *Environment Collection*: Gym includes a wide range of environments, from simple classical control tasks (like CartPole and MountainCar) to more complex robotics simulations. These environments are designed to test various aspects of RL algorithms.
- *Interface*: Gym provides a consistent API for interacting with different environments. This API includes methods like `reset()`, `step(action)`, and `render()`, allowing users to start an episode, apply actions, and view the environment, respectively.
- *Custom Environments*: Users can create their own environments by following Gym's API guidelines. This flexibility allows researchers to test RL algorithms on custom scenarios.

Table 7 Key Aspects of D4RL

Aspect	Description
Diverse Datasets	D4RL includes various types of datasets collected from different environments. These datasets typically contain trajectories from policies that are either suboptimal or generated through different levels of exploration.
Benchmark Environments	It encompasses a range of benchmark environments, including: • Classic Control: Simple environments like CartPole and MountainCar. • Mujoco: More complex robotic control tasks. • Adroit: Tasks involving dexterous manipulation using simulated robotic hands. • Robotic Manipulation: Scenarios involving more complex interactions with objects.
Offline RL	D4RL is particularly useful for offline RL, where the goal is to learn from a fixed dataset of trajectories without further interaction with the environment.
Metrics	It provides specific metrics to evaluate the performance of RL algorithms, such as Average Return and Success Rate.
Comparison	By using D4RL, researchers can compare different RL algorithms under the same conditions, ensuring a fair evaluation of their performance.

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- *Benchmarks*: Gym environments serve as benchmarks for comparing the performance of different RL algorithms, providing a common ground for evaluating progress in the field.
- *Integration*: Gym integrates with various RL libraries and frameworks, making it easy to use with popular RL toolkits like Stable Baselines, RLlib, and TensorFlow.

D4RL uses environments from OpenAI Gym as part of its benchmarking suite. The D4RL benchmark includes several classic control and robotics environments that are part of the Gym collection. These environments are used to create datasets for offline RL tasks and are integral to evaluating and comparing different RL algorithms. Some specific Gym environments used in D4RL include:

- *CartPole*: A classic control task where the goal is to balance a pole on a cart.
- *MountainCar*: A task where the agent must build up momentum to drive a car up a hill.
- *Acrobot*: A task involving a two-link robotic arm where the goal is to swing the end of the arm above a certain height.
- *LunarLander*: A simulation of landing a spacecraft on the moon.

7.1.2 MuJoCo

MuJoCo (Multi-Joint dynamics with Contact) [126] is a physics engine developed by Emo Todorov at the University of Washington. It is designed for high-fidelity simulation of robotics and biomechanics. Key features of MuJoCo include:

- *Realistic Physics Simulation*: MuJoCo provides accurate simulation of rigid body dynamics, contact forces, and soft constraints. It is well-suited for tasks that require high precision in physical interactions.
- *Continuous Control*: The engine is particularly useful for continuous control tasks, such as those involving complex robotic movements and manipulation. It supports simulations with a high degree of realism and detail.
- *Performance*: MuJoCo is known for its efficiency and speed, allowing for the simulation of complex environments with many interacting objects in real-time.
- *Visualization*: It includes tools for visualizing simulations, making it easier to debug and understand the behavior of robotic systems and other physical simulations.
- *Integration with Gym*: MuJoCo can be used with OpenAI Gym through the `mujoco-py` library. This integration allows users to access MuJoCo's high-fidelity simulations within the Gym framework, combining Gym's easy-to-use API with MuJoCo's advanced physics.

MuJoCo environments are also a significant part of D4RL, particularly for more complex and high-dimensional tasks. The D4RL benchmark includes environments from MuJoCo for evaluating offline RL performance in realistic and challenging scenarios. Some examples include:

- *HalfCheetah*: A 2D simulation of a robotic cheetah where the goal is to run as fast as possible.
- *Hopper*: A single-legged robot that hops across a flat surface.
- *Walker2d*: A 2D bipedal robot walking across a surface.
- *Ant*: A 3D quadrupedal robot navigating a simulated environment.
- *Humanoid*: A 3D humanoid robot walking or performing movements.

These MuJoCo environments are used in D4RL to provide high-fidelity simulations and complex tasks for offline RL algorithms.

7.1.3 Role in D4RL

In D4RL, both Gym and MuJoCo environments are used to:

- *Create Diverse Datasets*: Different environments provide datasets of various types, such as classical control tasks, complex robotic manipulations, and high-dimensional control problems.
- *Benchmark Algorithms*: The standardization of environments allows for fair and consistent benchmarking of RL algorithms across different tasks and difficulty levels.
- *Evaluate Offline Learning*: The environments help in assessing how well offline RL algorithms can perform given a fixed dataset, without further interaction with the environment.

Table 8 summarizes the environment used in offline RL.

7.2 Enhancing offline RL datasets

Despite the availability of various offline RL datasets for autonomous driving, significant challenges persist, particularly concerning static data limitations and distribution biases. Several recent studies [143, 144] have explored potential improvements to existing datasets, which are discussed in the following.

- *Augmented data generation for enhanced policy learning:* A critical limitation of real-world datasets is their incompleteness, particularly regarding action and reward information. To address this, Fang et al. [143] introduced a data-driven simulator that augments real-world datasets with synthetic yet realistic interactions. By simulating diverse driving scenarios based on the INTERACTION dataset [145], this approach enables the generation of additional training samples that better capture edge cases and improve policy robustness.
- *Combining expert and exploratory data:* Traditional datasets primarily contain expert demonstrations, which may not adequately expose RL algorithms to a diverse set of state-action distributions. Hao et al. [144] showed that incorporating exploratory data—such as randomly generated or failure scenarios—into training datasets significantly improves offline RL performance. Their study demonstrates that policies trained on mixed datasets (expert + exploratory) achieve better generalization and robustness compared to those trained exclusively on expert data.
- *Multi-source data integration to mitigate biases:* Single-source datasets often introduce biases due to their specific data collection environments, leading to poor generalization in unseen conditions. To overcome this, leveraging multiple real-world datasets, such as INTERACTION and Argoverse [146] can provide a more comprehensive state-action space coverage. Integrating heterogeneous datasets enables offline RL models to learn from a richer variety of driving behaviors and environmental conditions, reducing overfitting to a particular driving style or road structure.

By incorporating these strategies, offline RL datasets can be significantly improved, allowing for more robust policy evaluation and training.

8 Evaluation metrics in offline RL

In offline RL, evaluating the performance of learned policies is critical to understanding their effectiveness and robustness. Here are some commonly used evaluation metrics and techniques in offline RL:

1. *Return Metrics:* The first metric is Average Return, which represents the average cumulative reward a policy receives over a series of episodes. It is calculated as (14):

$$AverageReturn = \frac{1}{N}\sum_{i=1}^{N} R_i \tag{14}$$

Table 8 Environments used for evaluating offline RL algorithms

Environment	Library	Description	State space	Action space	Reward	Refs.
CartPole	Gym	A classic control problem where the goal is to balance a pole on a cart by applying forces to the cart.	Position, velocity, angle, angular velocity	Discrete (left, right)	+1 for every step the pole remains upright	[127]
Acrobot	Gym	A two-link pendulum environment where the goal is to swing the end of the lower link to a given height.	Angular position and velocity of two links	Discrete (torque applied to joint)	−1 for every step until the goal is reached	[128]
LunarLander	Gym	A rocket trajectory optimization problem where the goal is to land a lunar module on a landing pad.	Position, velocity, angle, angular velocity, leg contact	Discrete (main engine, left/right engines)	+100 for landing, −100 for crashing, +10 for each leg contact, −0.3 for each engine firing	[129]
HalfCheetah	Mujoco	A 2D robot with 9 links and 8 joints, where the goal is to make the cheetah run forward as fast as possible.	Position and velocity of 9 links	Continuous (torques applied to joints)	Based on the distance moved forward	[130]
Hopper	Mujoco	A single-legged robot that hops forward, with the goal of maximizing forward distance.	Position and velocity of 4 body parts	Continuous (torques applied to joints)	Based on the distance hopped	[131]
Walker2d	Mujoco	A two-legged robot that walks forward, with the goal of maximizing forward distance.	Position and velocity of 6 body parts	Continuous (torques applied to joints)	Based on the distance walked	[132]
Ant	Mujoco	A four-legged robot that walks forward, with the goal of maximizing forward distance.	Position and velocity of 13 body parts	Continuous (torques applied to joints)	Based on the distance moved forward	[133]
Humanoid	Mujoco	A humanoid robot that walks forward, with the goal of maximizing forward distance.	Position and velocity of 17 body parts	Continuous (torques applied to joints)	Based on the distance walked	[134]
Four Rooms domain	Gym	A grid-based environment with four rooms connected by corridors, where the goal is to navigate to a target location.	Position in a grid	Discrete (up, down, left, right)	Based on reaching the goal	[5]
AntMaze	Mujoco	A maze environment where an ant robot must navigate to a target location.	Position and velocity of 13 body parts	Continuous (torques applied to joints)	Based on reaching the goal	[135]
Adroit	Mujoco	A dexterous hand manipulation environment with various tasks such as object relocation and in-hand manipulation.	Position and velocity of 24 body parts	Continuous (torques applied to joints)	Based on task-specific goals	[136]
Kitchen	Mujoco	A kitchen environment where a robot must perform various tasks such as opening doors and manipulating objects.	Position and velocity of various kitchen objects	Continuous (torques applied to joints)	Based on task-specific goals	[137]
Atari games	Gym	A collection of classic Atari games used for RL research.	Pixel values of the game screen	Discrete (game-specific actions)	Game-specific scoring	[138]
walker-walk	Mujoco	Similar to Walker2d, but with different task specifications.	Position and velocity of 6 body parts	Continuous (torques applied to joints)	Based on the distance walked	[132]
sawyer-door	Mujoco	A robotic manipulation task where a Sawyer robot must open a door.	Position and velocity of the robot and door	Continuous (torques applied to joints)	Based on opening the door	[139]
Locked Doors	Gym	A grid-based environment with locked doors that must be navigated to reach a target location.	Position in a grid	Discrete (up, down, left, right)	Based on reaching the goal	[5]
Procgen Mazes	Gym	A procedurally generated maze environment where the goal is to navigate to a target location.	Position in a procedurally generated maze	Discrete (up, down, left, right)	Based on reaching the goal	[140]
Quadruped	Mujoco	A four-legged robot that walks forward, similar to the Ant environment.	Position and velocity of 13 body parts	Continuous (torques applied to joints)	Based on the distance moved forward	[141]

Table 8 (continued)

Environment	Library	Description	State space	Action space	Reward	Refs.
Autonomous vehicle environments highway-env	Separate	Environments for testing autonomous vehicle algorithms, including highway driving scenarios.	Position, velocity, and other state variables of the vehicle and environment	Continuous (steering, acceleration, braking)	Based on task-specific goals	[142]

where R_i is the return of episode i and N is the total number of episodes. Another important metric is Median Return, which represents the middle value of returns across episodes. This measure is more robust to outliers than the average return and can provide a clearer picture of typical performance.

2. *Value-Based Metrics:* The Q-Value or Action-Value Function estimates the expected return when an action is taken in a given state. To evaluate this, one common metric is the Mean Squared Error between the estimated Q-values and the true Q-values, which helps in understanding how accurate the value predictions are.
3. *Distribution of States:* State Distribution Matching is a critical metric to assess how well the learned policy matches the state distribution present in the offline dataset. This ensures that the policy operates within the range of the data used during training. One way to measure this is through the KL Divergence, which quantifies the difference between the state distributions of the policy and the offline dataset. The formula is (15):

$$D_{KL}(\pi_{\text{data}} \parallel \pi_{\text{policy}}) = \sum_{s \in \mathcal{S}} p_{\text{data}}(s) \log \frac{p_{\text{data}}(s)}{p_{\text{policy}}(s)} \tag{15}$$

where $p_{\text{data}}(s)$ is the state distribution from the offline dataset, and $p_{\text{policy}}(s)$ is the state distribution of the learned policy.

4. *Stability Across Random Seeds:* Stability Across Random Seeds evaluates the consistency of a policy's performance across different random seeds, which is essential for understanding the robustness and reliability of the model. Key metrics include the Standard Deviation of Returns, which measures the variability of the policy's performance across random seeds, calculated as (16):

$$StdDev = \sqrt{\frac{1}{K} \sum_{k=1}^{K} \left(R_k - \bar{R}\right)^2} \tag{16}$$

where R_k is the return for the k-th random seed, and $\bar{R}$ is the mean return. Additionally, **Confidence Intervals** (17) provide a statistical measure of the policy's reliability, calculated as:

$$CI = \bar{R} \pm t_{\alpha/2} \cdot \frac{\text{StdDev}}{\sqrt{K}} \tag{17}$$

where $t_{\alpha/2}$ is the critical value for the desired confidence level.

5. *Off-Policy Evaluation:* Off-Policy Evaluation (OPE) assesses a method's expected performance using only offline data. OPE is crucial for tasks such as hyperparameter tuning, model fine-tuning, and deciding when to stop training to avoid overfitting. The DOPE benchmark provides a standardized framework for comparing OPE algorithms across a wide range of tasks, leveraging datasets that reflect realistic scenarios like human demonstrations and random exploration. The DOPE benchmark includes metrics such as Absolute Error, which prioritizes robustness to outliers, Regret@k, which measures the performance gap between estimated top policies and the actual best policy, and Rank Correlation, which assesses how accurately the OPE estimates match the true rankings of policies.

8.1 Analysis of quantitative results

Normalized Score is the primary metric used by Fujimoto and Gu [147] to evaluate the performance of each algorithm across different tasks (e.g., HalfCheetah, Hopper, and Walker2d) in various dataset settings such as Random, Medium, Medium-Expert, and Expert. The normalized score represents the model's performance in comparison to the optimal performance, with higher values indicating better performance. These scores are averaged over 5 seeds, and the standard deviation across seeds is also reported, providing insights into the stability of the methods. TD3+BC achieves performance comparable to Fisher-BRC and CQL across most tasks, with slight variations in different dataset settings. For instance, in the HalfCheetah-Expert task, TD3+BC scores 106.8, which is nearly identical to CQL's 105.7 and Fisher-BRC's 106.1. In the Hopper task, TD3+BC consistently outperforms CQL and Fisher-BRC in Medium and Expert settings, scoring 99.4 and 112.3, respectively. In Walker2d, TD3+BC outperforms CQL in all settings, achieving better scores in Medium and Expert datasets (79.5 and 105.7) compared to CQL's 57.5 and 103.8.

Evaluates the performance of different variations of CQL [12] showed that CQL(H) generally outperforms others in MuJoCo tasks, with significantly higher returns in environments like halfcheetah-medium-expert (7234.5 vs. 3995.6). Including the dataset maximization term improves performance, as seen in the hopper-medium task (1866.1 vs. 1028.4 without it). Additionally, CQL(H)-Lagrange, which adapts the regularization weight dynamically, outperforms manually tuned versions in complex tasks such as AntMaze-medium-diverse (0.53 vs. 0.21), demonstrating its robustness in challenging scenarios. However, the adaptive version provides less benefit in simpler MuJoCo tasks.

MABE [110] performs well, achieving the top score in 7 out of 9 D4RL datasets compared to previous methods. It consistently outperforms MOPO, MOReL, SAC, CQL, and BRAC-v in several environments. On average, MABE achieves a score of 77.5, while CQL scores 63.9 and MOPO scores 72.9. This shows that MABE performs well across different tasks and environments.

BooT [32] achieves a 16.8% improvement in average score and outperforms other methods across all 9 datasets. BooT performs with an average score of 81.9, slightly ahead of other methods like BooT-r, AR (79.1) and BooT-o, AR (79.9), reflecting the overall strength of BooT in both trajectory generation and bootstrapping.

COMBO [50] demonstrates strong performance across various D4RL datasets, consistently outperforming other offline RL algorithms such as MOPO, MOReL, CQL, SAC-off, BEAR, BRAC-p, and BRAC-v. COMBO consistently outperforms BC and other baselines across a wide range of tasks, with an average score of 77.5, compared to CQL's 63.9 and MOPO's 72.9. This shows the effectiveness and adaptability of COMBO in both simple and complex environments.

In more recent study, the averaged normalized scores was used to evaluate the performance of the POR [99] compared to other baseline algorithms across various tasks from the D4RL dataset. The results are averaged over the final 10 evaluations with 5 different seeds to ensure robustness. POR achieved the highest score in 11 out of 18 tasks, demonstrating its competitive edge over the other baselines. For example, in the antmaze-u task, POR scored 90.6, outperforming the best baseline BCQ, which scored 89.0. Similarly, in the antmaze-l-d task, POR achieved 73.4, significantly surpassing the best baseline CQL, which scored 45.6.

8.2 Computational efficiency

Computational efficiency plays a key role in the practical deployment of offline RL methods, particularly when training models on large-scale datasets. However, most studies in this field provide limited or inconsistent information on runtime and memory usage, making direct comparisons challenging.

Among the few works that report training time, TD3+BC is noted for slightly shorter training durations compared with methods such as Fisher-BRC and CQL (e.g., approximately 2 h 8 min vs. 2 h 12 min and 4 h 11 min, respectively). While these results suggest modest improvements in efficiency, the differences are relatively small and may be influenced by experimental setup, hardware configurations, or hyperparameter tuning rather than inherent algorithmic advantages.

Comprehensive evaluations of computational efficiency remain scarce in the offline RL literature. Most studies prioritize performance metrics such as return scores or stability, while resource-related metrics such as runtime and memory are often underreported. Future research should therefore include standardized benchmarks for computational efficiency to provide a more complete understanding of trade-offs between performance and computational cost in offline RL methods.

9 Discussion

Offline RL has significant potential in high-stakes domains such as autonomous driving and robotics, where real-world interactions are costly, risky, or impractical. In autonomous driving, vehicles must make decisions based on previously collected data, often from a variety of environments (e.g., urban areas, highways, adverse weather conditions). Offline RL is particularly relevant here, as the algorithm can learn policies from this static data without requiring further real-world interactions, which could be expensive or unsafe. One critical challenge is handling scenarios that deviate significantly from the training data distribution, such as encountering an unfamiliar road layout or extreme weather conditions.

Historically, model-free offline RL algorithms have depended on policy constraints [53, 147] to prevent the learned policy from generating OOD actions. These constraints have taken various forms, including the addition of BC loss to measure divergence between the behavior and learned policies, the application of advantage-weighted constraints, penalization of prediction errors from variational auto-encoders [97], and the learning of latent actions from offline data. However, these approaches may result in overly conservative value functions and heavy reliance on the behavior policy, despite some OOD actions potentially being reliable if they closely match the offline data's support. CQL addresses this by minimizing Q-values for OOD samples, thus implicitly constraining the policy.

Model-based approaches like MOPO and MOReL utilize ensembles of dynamics models for uncertainty quantification and apply pessimistic updates to the value function. Unlike earlier work that applies conservatism directly to state-action pairs or actions, these approaches focuses on state-level conservative estimation without the need for explicit uncertainty quantification. CODAC builds on Q-value restriction methods, but by acquiring conservative quantifications for the entire spectrum of quantile values within the return distribution, instead of merely focusing on the expected return. On the other hand, the other risk-sensetive algorithm, 1R2R, avoids combining different techniques, shows that solely being risk-averse can sufficiently prevent distributional shift without the need for a computationally intensive distributional value function.

Integrating RL with BC and imitation learning is also a notable trend in offline RL, aimed at enhancing learning from existing datasets. This strategy is implemented in algorithms like DDPG [27] and natural policy gradient [148], boosting learning speed, exploration, and efficiency. Notable examples include the integration of Soft Actor-Critic (SAC) with BC [47] and the combination of Proximal Policy Optimization (PPO) with BC [149]. Additionally, [48] modified the TD3 algorithm for offline use by adding a BC term to the policy update step, significantly reducing computational demands.

A challenge arises as most methods apply a uniform constraint on the policy, ignoring the varying degrees of OOD actions. To address this, uncertainty estimation techniques have been introduced. For instance, UWAC employs MC-dropout [150] for local uncertainty estimation but neglects the broader uncertainty across future trajectories, a gap in the literature for policy-constraint methods. Q-restriction approaches like SAC-N [75], MOPO, and PBRL consider local uncertainty as an intrinsic reward, incorporating it into the Q-value estimation. PBRL, in particular, distinguishes itself by explicitly quantifying uncertainty for OOD actions, in contrast to CQL's uniform penalization. In contrast to MOReL, which creates terminal states using a fixed uncertainty threshold,

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MOPO applies a flexible reward penalty to integrate uncertainty into its framework. ABM learns an adaptive behavior prior similar to MABE, but learns the policy using the model-free MPO algorithm.

Uncertainty-based methods in offline RL, borrowing from online RL strategies like bootstrapped Q-networks and ensemble dynamics, face additional challenges due to limited data coverage and policy distribution shifts. While BOPAH combines uncertainty penalization with behavior-policy constraints, UWAC and PBRL adopt dropout-based uncertainty estimations without the need for further policy constraints, focusing on penalizing OOD actions through direct sampling and uncertainty quantification. PBRL, proposes a novel approach of explicit value underestimation for OOD actions based on uncertainty, requiring fewer ensemble networks compared to methods like SAC-N, which necessitates a large ensemble for Q-value estimation. COMBO offers a more cautious and precise value approximation compared to CQL, while CSVE reaches equivalent minimum value estimates as COMBO across a broader range of state distributions. On the other hand, EDAC computes the gradient of every Q-function and ensures variation in these gradients to adequately penalize actions that are out of distribution. Meanwhile, UTDS imposes penalties on out-of-distribution actions by directly sampling such actions and quantifying the related uncertainty.

While CQL aims to establish a lower bound for the policy's value function by minimizing Q-values for OOD actions, there's a noted lack of analysis on the error propagation in policy iteration within offline RL contexts. The PPI method seeks to fill this gap by exploring the sub-optimality in offline policy iteration and adjusting constraints based on policy uncertainty, potentially uncovering viable actions close to the dataset's support. This adaptability, aligned with uncertainty estimation in offline RL, suggests a move towards more flexible and efficient approaches in managing OOD actions and policy generalization within the constraints of offline RL datasets. The integration of uncertainty estimation presents a promising avenue for enhancing policy robustness and generalization.

Several researches have studied the adoption of pessimistic approaches to mitigate the limitations of overly optimistic value estimations. By assuming worst-case scenarios for unknown states or actions, these methods aim to develop more reliable, albeit conservative, policies. This approach often leads to more reliable but potentially more conservative policies. Approaches that estimate a conservative or pessimistic value function might avoid overestimation but can suffer from underestimation. Excessive pessimism in conservative value function methods is a key issue, and how to modulate conservatism dynamically to balance the risks of overestimation with the drawbacks of avoiding unfamiliar actions.

Regularization in offline RL is employed to enhance the learned policy by including a penalty term in the objective function. Policy regularization modifies the policy gradient objective to not only maximize the expected return but also a regularization term. Table 9 presents a comparative analysis of various approaches, highlighting the advantages and limitations of each.

10 Challenges and future direction

Despite the considerable progress made in offline RL, several significant challenges remain that must be addressed for more widespread and effective application. These challenges are crucial to overcome to ensure the robustness and adaptability of offline RL systems:

- *Out-of-Distribution:* A persistent challenge in offline RL is ensuring reliable performance when encountering unseen states or actions outside the training distribution. Despite various approaches, OOD issues remain a key bottleneck that limits real-world applicability. Models trained solely on static datasets often fail to generalize under distribution shifts, leading to unsafe or suboptimal behavior. Future work should therefore continue to focus on developing uncertainty modeling to mitigate OOD risks.
- *Model Overfitting:* In offline RL, there is a risk that models overfit to the training data, particularly when the offline dataset does not cover the full spectrum of states and actions that the agent may encounter. This leads

Table 9 Comparison of different approaches for handling distribution shift in Offline RL

Approach	Pros	Cons
Q-Restriction	- Guarantees that the learned value function is a lower bound of its true value (CQL). - Handles OOD and extrapolation error well by minimizing Q-values for OOD samples (CQL) or V-function at state-level conservatism (CSVE). - Penalizes high-entropy policies with mild conservatism (MCQ). - Distinguishes ID from OOD actions and ensures unbiased estimation for ID actions (SVR). - Converges to optimal support-constrained policy and is less susceptible to model errors (SVR)	- Penalizes Q-values of all OOD samples equally. - Maintains a constant level of pessimism, ignoring the OOD degree, which may hurt generalization. - May underestimate or overestimate Q-values (CQL). - Difficulty in determining the OOD region. - Requires pre-training the behavior model (SVR)
Policy-Constraint	- Handles OOD by staying close to behavior policy. - Balances policy closeness with flexibility (Fisher-BRC). - Can use behavior model as a prior (ABM). - Uses BC initialization with trust-region updates rather than explicitly forcing policy closeness (BREMEN)	- Suppresses OOD actions that could be trustworthy. - May cause overly conservative value function. - Biased towards behavior policy. - Can propagate behavior policy errors. - Risky if behavior policy is not optimal. - Data might come from non-Markovian policies in the real world. - Lacks analysis on error propagation of policy iteration
Uncertainty-based	- Conducts explicit uncertainty quantification for OOD actions rather than penalizing all OOD samples equally. - Ensures sufficient penalization for OOD actions using ensemble Q-networks (EDAC). - Bootstrapped uncertainty is more robust to data shift than dropout-based approaches (PBRL). - Restricts Q-value error under the trained policy's distribution (PPI). - Can perform well even without uncertainty estimation and benefits from adaptive behavioral priors (MABE)	- Uncertainty quantification is more challenging in offline RL due to limited offline data coverage. - Performance is limited by the data distribution. - Uncertainty estimation is more limited in practical situations with deep neural networks. - Some approaches implicitly calculate global uncertainty using hundreds of Q-networks, which is computationally inefficient

to poor performance when the model faces distribution shifts that deviate from the training distribution. Overfitting is particularly concerning when models lack generalization, which can compromise decision-making in dynamic environments [91]. Future work could explore advanced regularization techniques or robust algorithms that improve generalization capabilities in the face of unseen data.

- *Complexity of the Environment:* Real-world environments are inherently dynamic and complex, often exhibiting continuous changes in data distribution. This makes it difficult to model distribution shifts accurately and adapt effectively. Furthermore, changes in the environment may occur unpredictably, adding additional challenges to the adaptation process [151]. Tackling this issue requires more advanced modeling techniques that can account for these dynamic shifts and provide real-time updates to offline models. Future approaches might explore contextual adaptation or dynamic model architectures to handle such environments more effectively.
- *Scalability:* As datasets grow larger and more complex, the ability of offline RL algorithms to scale efficiently becomes increasingly important. Many offline RL algorithms struggle with the computational complexity associated with large, high-dimensional datasets, and this limits their applicability in practical, real-world scenarios [152]. Future research should focus on computational efficiency and distributed learning frameworks that can manage massive datasets with varying degrees of distribution shifts. Techniques such as parallel processing or distributed optimization [153] could help address these scalability challenges.

As offline RL continues to evolve, there are several critical areas that require further exploration to address existing challenges and improve the robustness of algorithms in real-world applications. In this context, the following areas of research offer promising opportunities for advancing the field:

- *Synthetic Data Generation:* One promising area for future research is the generation of synthetic data that is more representative of real-world distribution shifts and OOD scenarios. Synthetic data could be used to create a more diverse range of states and actions, especially for environments where collecting diverse real-world data is impractical. This could involve techniques such as data augmentation or leveraging generative models like GANs to simulate a variety of OOD conditions that may not be present in the original offline dataset [154, 155].
- *Interpretable Models:* As offline RL models become more complex, understanding how they make decisions in the presence of distribution shifts is increasingly important. Designing interpretable models can provide valuable insights into how distribution shifts impact decision-making, helping researchers identify when and why models fail. Furthermore, interpretable models can guide the development of more reliable models by highlighting key decision-making factors that may be influenced by shifts, which is crucial for real-world deployment [156, 157].
- *Adaptive Algorithms:* One key area for future work is developing adaptive algorithms that can dynamically adjust to distribution shifts without requiring complete retraining. Techniques such as meta-learning or adaptive learning rates could be leveraged to create models that can learn to adapt more quickly to new data distributions. These approaches could improve both the efficiency and effectiveness of offline RL in real-world applications, particularly in environments that are subject to frequent or unpredictable changes [78, 158].
- *Human-in-the-Loop:* Integrating human feedback into the offline RL process is another promising avenue for addressing distribution shifts and OOD issues. By incorporating expert knowledge or real-time human input, offline RL models could more effectively identify and address the challenges posed by unseen data distributions. This approach could be particularly useful in complex, high-stakes environments, such as healthcare and autonomous driving, where expert intervention can provide critical insights into model behavior [159].
- *Transfer Learning:* The use of transfer learning techniques offers a way to leverage knowledge from related tasks or environments to address distribution shifts in offline RL scenarios. Transfer learning allows models to apply insights gained from one task to a new, but related task, potentially reducing the amount of data required to adapt to shifting distributions. Future work could explore multi-task learning or domain adaptation strategies to enhance the generalization capabilities of offline RL algorithms across a wider range of applications [110, 160].

11 Conclusion

The field of Offline RL is undergoing rapid evolution, with a strong focus on developing algorithms capable of addressing the inherent challenges of learning from fixed datasets. This study has explored the critical issues of distributional shifts and OOD actions, emphasizing the pivotal role of uncertainty quantification in enhancing the robustness and generalization of Offline RL policies.

Our analysis revealed that approaches based on Q-value restriction and policy constraints have demonstrated significant potential in mitigating the risks associated with distributional shifts. As we discussed, in both approaches, incorporating uncertainty quantification holds potential for improving the robustness and generalization of Offline RL policies.

Moreover, this study highlights the importance of aligning algorithmic development with practical benchmarks and evaluation metrics, as these are critical for translating theoretical advancements into real-world impact. The categorization of methods and the detailed analysis provided in this work aim to serve as a valuable resource for researchers and practitioners, facilitating informed decisions when selecting or designing Offline RL techniques (Table 10).

Table 10 Summary of symbols and parameters used in equations

Symbol	Description
s	State
a	Action
T	Transition probability function
d	Distribution
r	Reward
γ	Discount factor
o	Observation
E	Expectation (objective) function
π	Policy
τ	Trajectory
V	Value estimation function
θ	Parameters of the learned Q-function
$J(\theta)$	Bellman objective
π_β	Behavioral policy
$\mathcal{B}_\pi$	Empirical Bellman operator
c	Scaling factor for penalties
α	Coefficient adjusting penalty magnitude
$\mathcal{L}_p$	Loss function
F	Cumulative distribution function (CDF)
D	Dataset
u	Bellman uncertainty function
$V(s)$	State-value function
g	Guide policy
$\mu(s, a)$	State–action pair distribution
U	Uniform distribution $\mathcal{U}([0, 1])$

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Data availability This study is a survey paper and does not involve the generation or analysis of any new datasets. All datasets and works referenced are publicly available and cited accordingly throughout the manuscript.

Materials availability Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors and Affiliations

Ali Riahi Samani[1] · Xujiang Zhao[2] · Feng Chen[1]

✉ Ali Riahi Samani
ali.riahi@utdallas.edu

Xujiang Zhao
xuzhao@nec-labs.com

Feng Chen
feng.chen@utdallas.edu

[1] Department of Computer Science, The University of Texas at Dallas, Richardson, Texas, USA

[2] NEC Laboratories America, Princeton, New Jersey, USA